

```
In [72]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler, OneHotEncoder, MinMaxScaler
from sklearn.metrics import accuracy_score, precision_score, recall_score
from sklearn.metrics import classification_report, ConfusionMatrixDisplay
from sklearn.linear_model import LogisticRegression
from sklearn.compose import ColumnTransformer
from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.pipeline import Pipeline

from catboost import CatBoostClassifier
from xgboost import XGBClassifier

from imblearn.over_sampling import SMOTE
from imblearn.under_sampling import RandomUnderSampler
from imblearn.pipeline import Pipeline as imbPipeline

sns.set(style="whitegrid")
```

1. EDA

1.1 Data exploration

```
In [73]: df = pd.read_csv("/Users/adilkhankaldybekov/Desktop/adil Khan/Machine Learning/credit_data.csv")
```

```
In [74]: #Data Set observation
```

```
print(df.shape)
df.head()
```

(45211, 17)

```
Out[74]:
```

	age	job	marital	education	default	balance	housing	loan	contact
0	58	management	married	tertiary	no	2143	yes	no	unknown
1	44	technician	single	secondary	no	29	yes	no	unknown
2	33	entrepreneur	married	secondary	no	2	yes	yes	unknown
3	47	blue-collar	married	unknown	no	1506	yes	no	unknown
4	33	unknown	single	unknown	no	1	no	no	unknown

```
In [75]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45211 entries, 0 to 45210
Data columns (total 17 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         45211 non-null  int64
1   job         45211 non-null  object
2   marital     45211 non-null  object
3   education   45211 non-null  object
4   default     45211 non-null  object
5   balance     45211 non-null  int64
6   housing     45211 non-null  object
7   loan        45211 non-null  object
8   contact     45211 non-null  object
9   day         45211 non-null  int64
10  month       45211 non-null  object
11  duration    45211 non-null  int64
12  campaign    45211 non-null  int64
13  pdays      45211 non-null  int64
14  previous    45211 non-null  int64
15  poutcome    45211 non-null  object
16  y           45211 non-null  object
dtypes: int64(7), object(10)
memory usage: 5.9+ MB
```

In [76]: *#Counting number of nan values for each feature*

```
print(df.isna().sum())
```

```
age         0
job         0
marital     0
education   0
default     0
balance     0
housing     0
loan        0
contact     0
day         0
month       0
duration    0
campaign    0
pdays      0
previous    0
poutcome    0
y           0
dtype: int64
```

In [77]: *#Data observation for integer features*

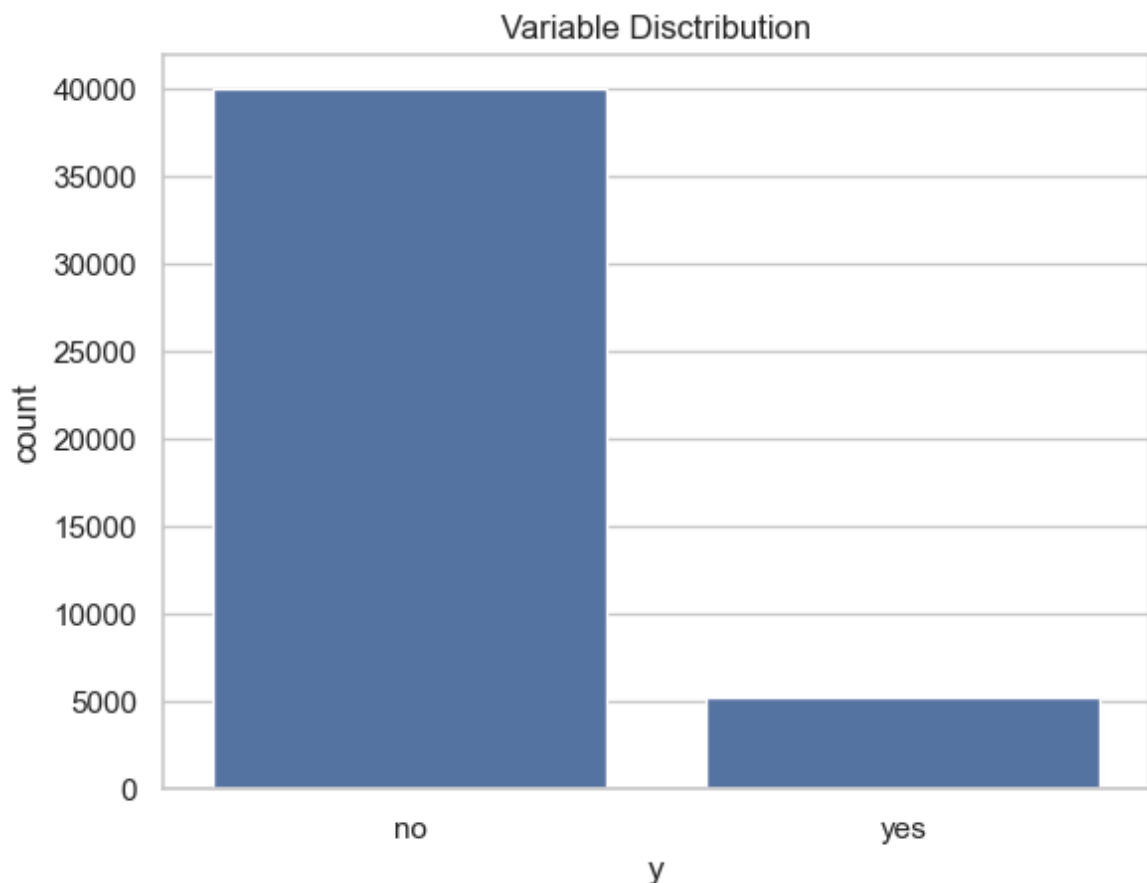
```
num_cols = df.select_dtypes(include=['int64', 'float64']).columns
df[num_cols].describe()
```

Out [77]:

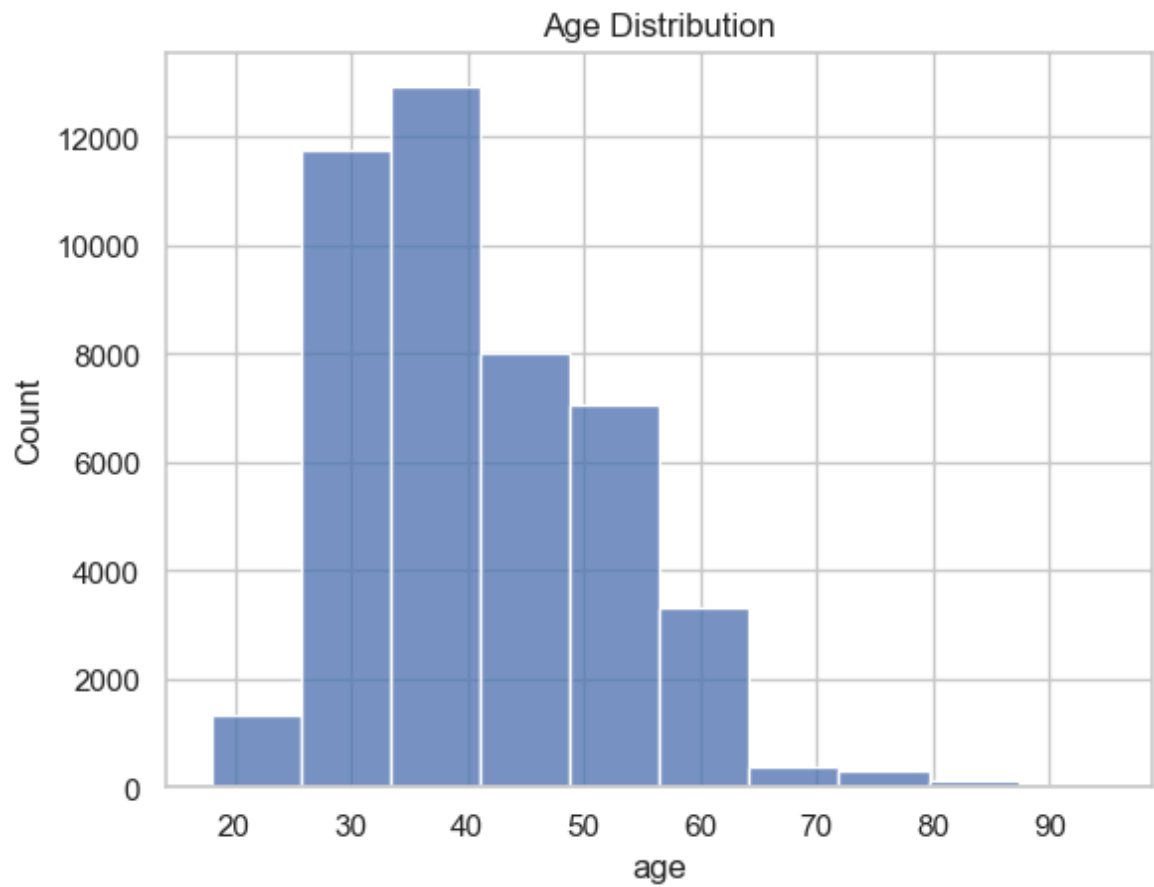
	age	balance	day	duration	campaign	
count	45211.000000	45211.000000	45211.000000	45211.000000	45211.000000	45211.000000
mean	40.936210	1362.272058	15.806419	258.163080	2.763841	2.763841
std	10.618762	3044.765829	8.322476	257.527812	3.098021	3.098021
min	18.000000	-8019.000000	1.000000	0.000000	1.000000	1.000000
25%	33.000000	72.000000	8.000000	103.000000	1.000000	1.000000
50%	39.000000	448.000000	16.000000	180.000000	2.000000	2.000000
75%	48.000000	1428.000000	21.000000	319.000000	3.000000	3.000000
max	95.000000	102127.000000	31.000000	4918.000000	63.000000	63.000000

1.2 Data visualization

```
In [78]: #Distribution of all target values in dataset
sns.countplot(x='y', data=df)
plt.title('Variable Distribution')
plt.show()
```

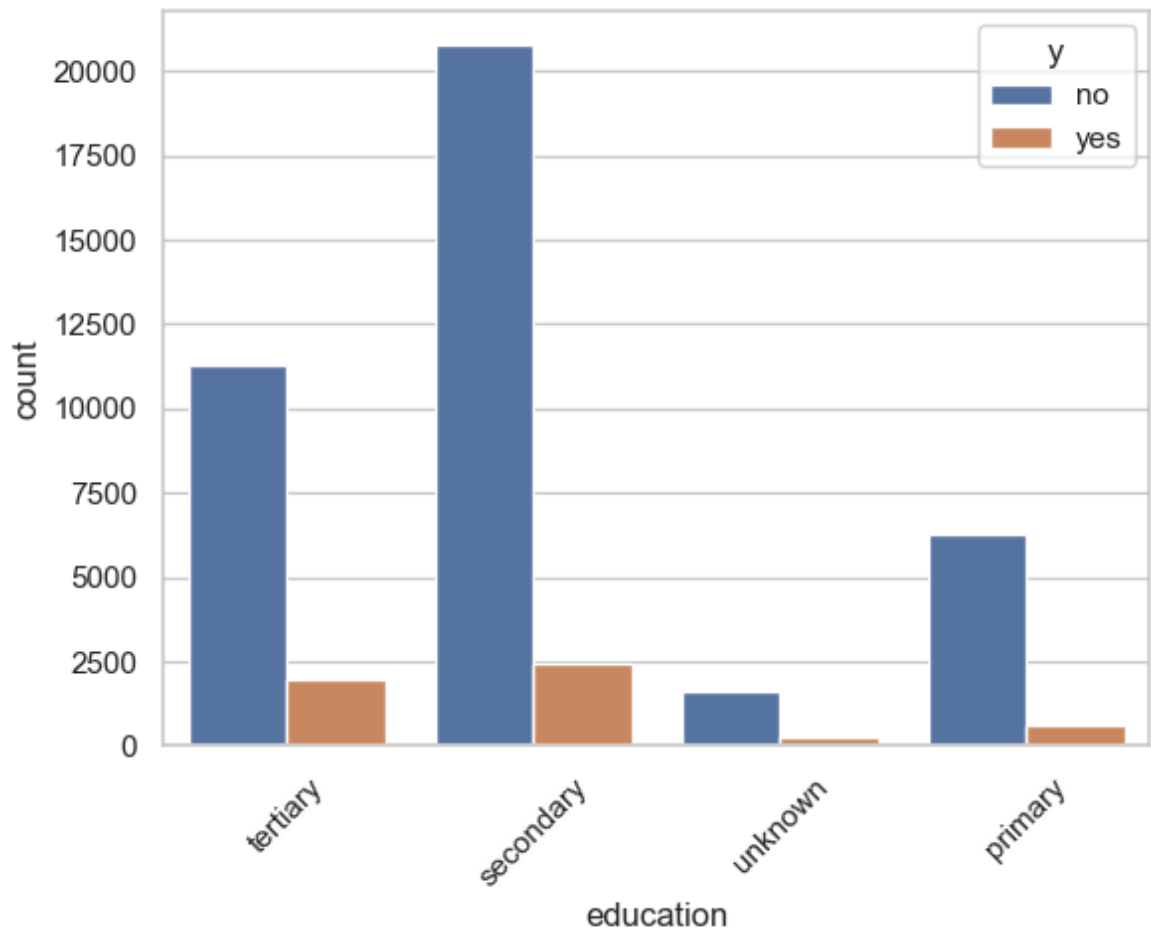


```
In [79]: # Visualizing the distribution of 'age' column
sns.histplot(df['age'], bins=10)
plt.title('Age Distribution')
plt.show()
```

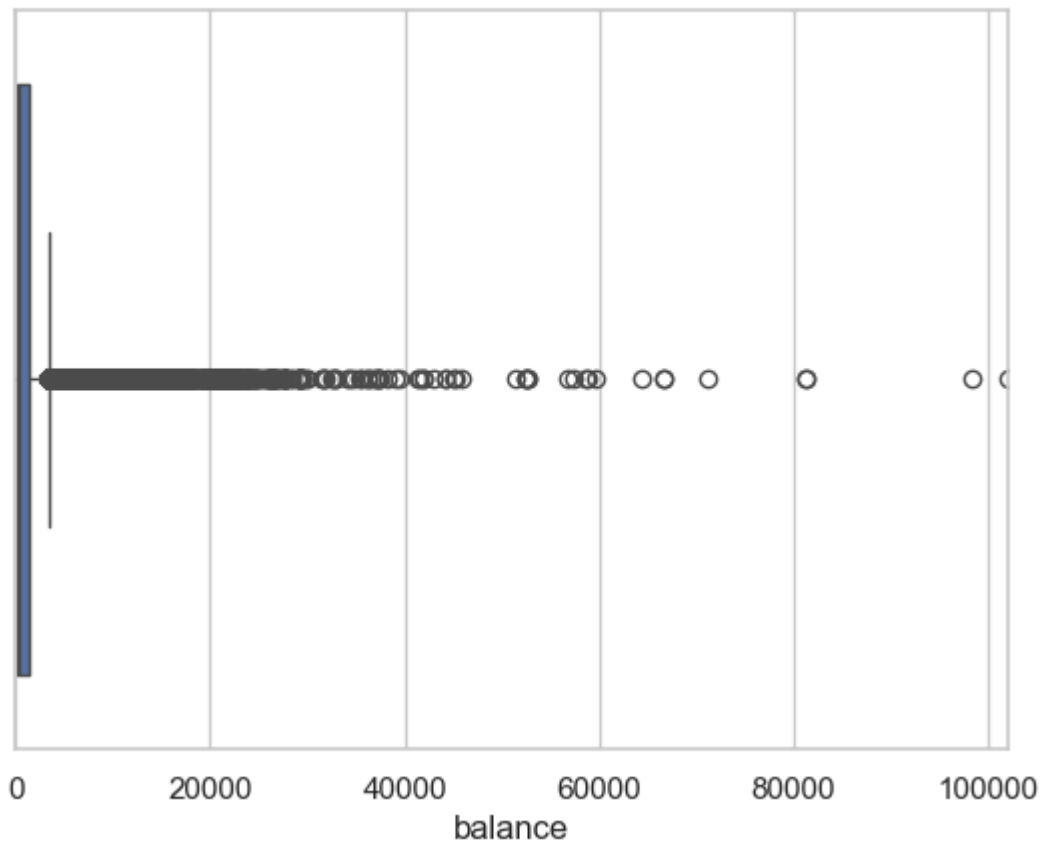


In [80]: *#Comparing of all education levels by target value*

```
sns.countplot(x='education', hue='y', data=df)
plt.xticks(rotation=45)
plt.show()
```

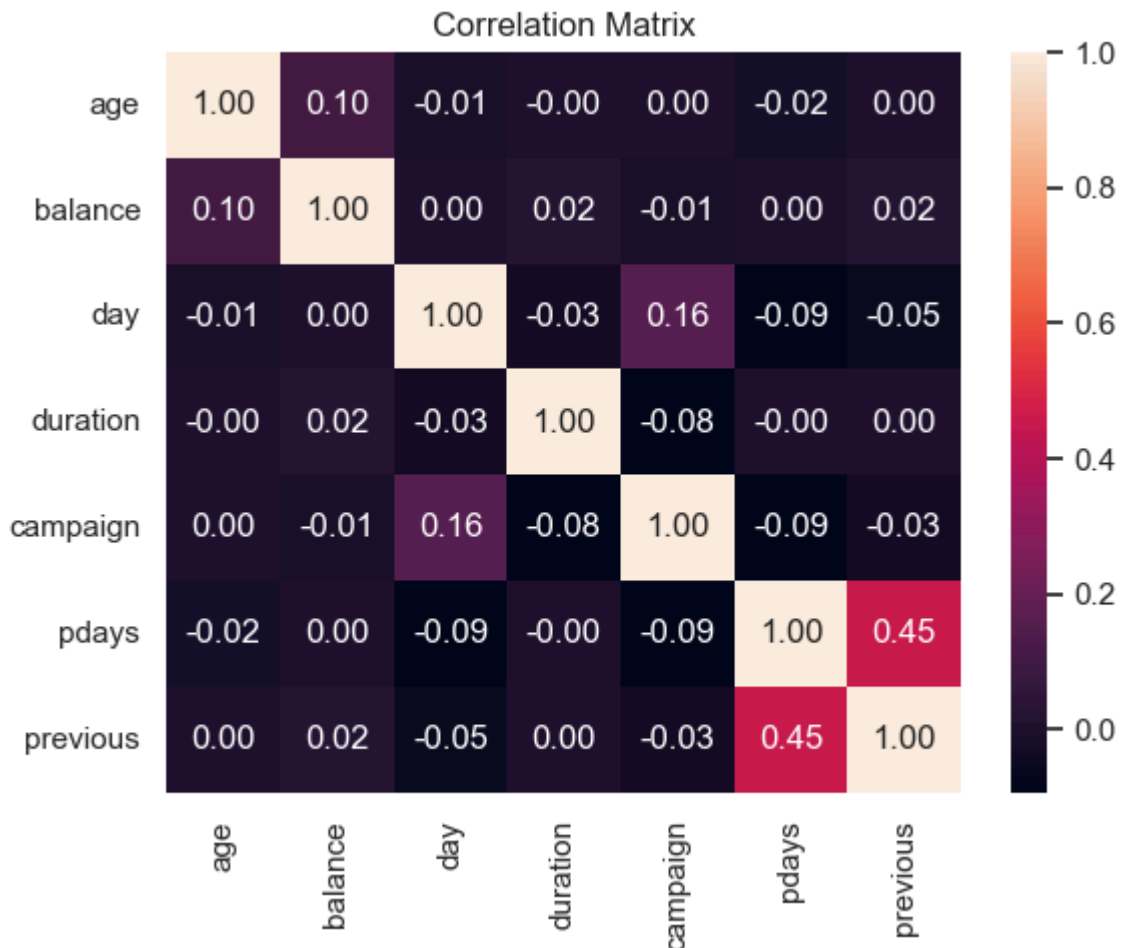


```
In [81]: sns.boxplot(x=df['balance'])  
plt.xlim(0, df['balance'].max())  
plt.show()
```



1.3 Correlation Analysis

```
In [82]: #looking at the relation between numerical features
correlation = df[num_cols].corr()
sns.heatmap(correlation, annot=True, fmt='.2f')
plt.title('Correlation Matrix')
plt.show()
```



2. Data processing/engineering

```
In [83]: cat_cols = df.select_dtypes(include=['object']).drop(columns=['y']).columns
```

```
In [84]: def create_prep():
    num_transformer = Pipeline(steps=[('scaler', StandardScaler())]) #for
    cat_transformer = Pipeline(steps=[('onehot', OneHotEncoder(handle_unknown='ignore'))])
    prep = ColumnTransformer(transformers=[('num', num_transformer, num_cols), ('cat', cat_transformer, cat_cols)])
    return prep

preprocessor = create_prep()
```

According to some observations we did data exploratory analysis and feature engineering engineering(Standardize data and Encoding)

2. Model Building (Построение моделей)

3.1 Logistic Regression

In [85]: `X = df.drop(columns='y', axis=1)`

```
Y = (df['y'] == 'yes').astype(int)
#Y = df['y'].map({'yes': 1, 'no' : 0})
```

In [86]: `Xtrain, Xtest, Ytrain, Ytest = train_test_split( X, Y, test_size=0.2, ran`

In [87]:

```
pipe = imbPipeline(steps=[
    ('prep', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('logreg', LogisticRegression(random_state=42, max_iter=1000))
])

pipe.fit(Xtrain, Ytrain)
Ypred = pipe.predict(Xtest)
Yprob = pipe.predict_proba(Xtest)[:, 1]

print("Distribution of values after SMOTE:")
Xsmote, Ysmote = pipe.named_steps['smote'].fit_resample(preprocessor.fit_
print(Ysmote.value_counts())
```

Distribution of values after SMOTE:

```
y
0    31937
1    31937
Name: count, dtype: int64
```

Evaluating

In [88]: `from sklearn.metrics import roc_auc_score`

```
print("====Classification Report====")
print(classification_report(Ytest, Ypred, target_names=['No', 'Yes']))
print(f"ROC AUC : {roc_auc_score(Ytest, Yprob):.2f}")
```

```
====Classification Report====
               precision    recall  f1-score   support

      No           0.97         0.85         0.91         7985
      Yes           0.42         0.80         0.55         1058

   accuracy                   0.85         9043
  macro avg           0.69         0.83         0.73         9043
 weighted avg           0.91         0.85         0.87         9043
```

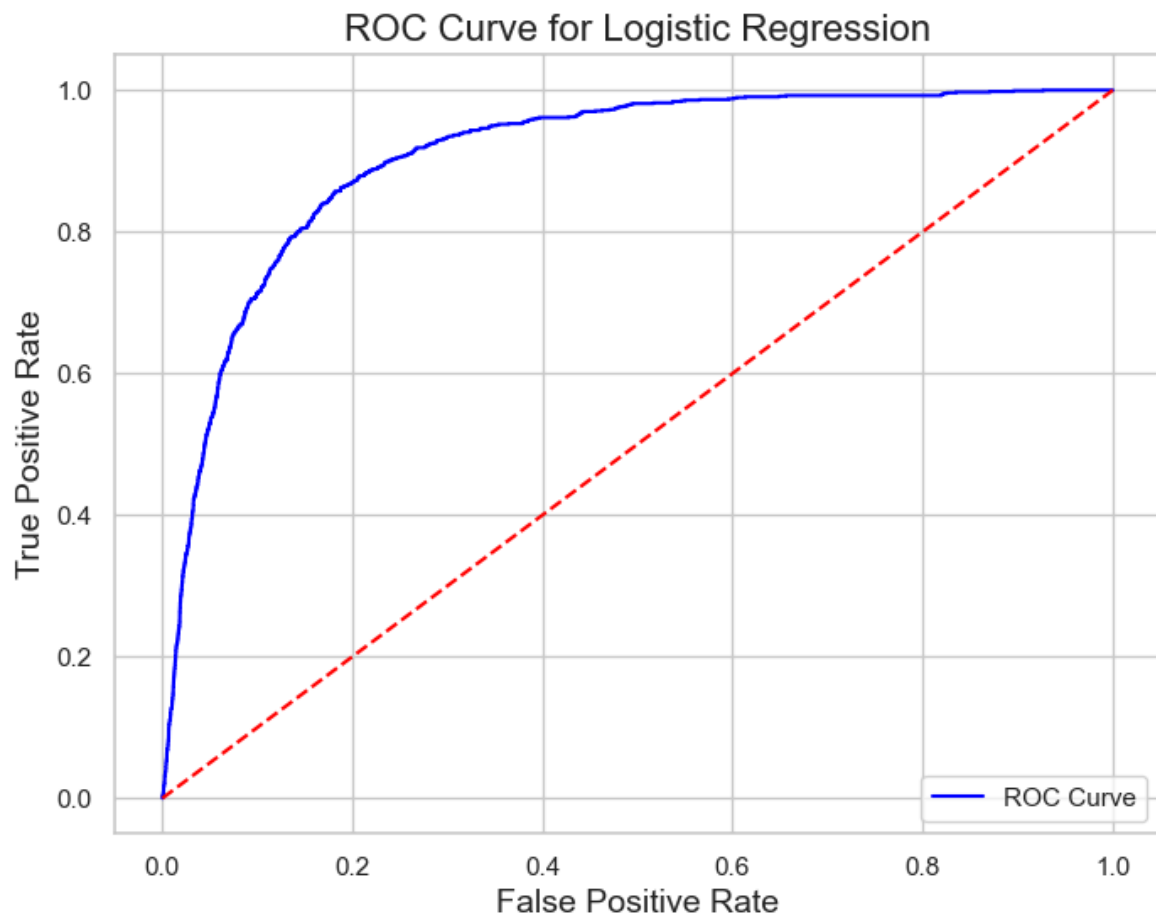
ROC AUC : 0.91

In [89]: `from sklearn.metrics import roc_curve`

```
fpr, tpr, thresholds = roc_curve(Ytest, Yprob)
```

```
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, label='ROC Curve', color='blue')
plt.title('ROC Curve for Logistic Regression', fontsize=16)
plt.xlabel('False Positive Rate', fontsize=14)
plt.ylabel('True Positive Rate', fontsize=14)
plt.legend(loc='lower right')
plt.plot([0, 1], [0, 1], linestyle='--', color='red')
```

Out[89]: [



Model Evaluation:

- **Accuracy:** Good overall, predicts the majority class well
- **Precision:** Low - many false positives
- **Recall:** Good due to SMOTE
- **ROC AUC:** Normal, but not high

3.2 Decision Tree

```
In [90]: pipe = imbPipeline(steps=[
    ('prep', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('DecTrees', DecisionTreeClassifier(random_state=42))
])

pipe.fit(Xtrain, Ytrain)
Ypred = pipe.predict(Xtest)
Yprob = pipe.predict_proba(Xtest)[:, 1]
```



```
In [91]: from sklearn.metrics import roc_auc_score

print("                ===Classification Report===")
print(classification_report(Ytest, Ypred, target_names=['No', 'Yes']))
print(f"ROC AUC : {roc_auc_score(Ytest, Yprob):.2f}")
```

```
                ===Classification Report===
           precision    recall  f1-score   support

     No         0.94         0.91         0.92         7985
     Yes         0.44         0.54         0.48         1058

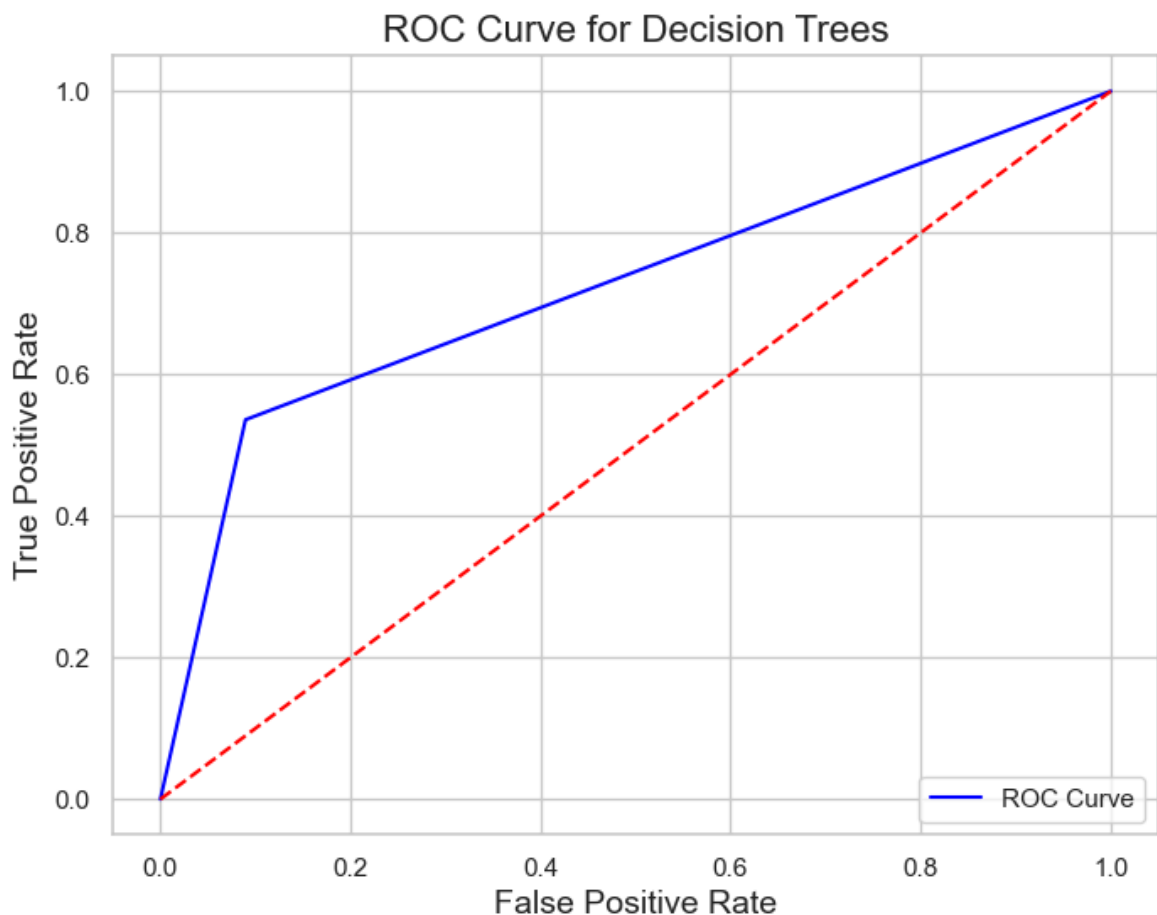
 accuracy          0.87         9043
 macro avg         0.69         0.72         0.70         9043
 weighted avg      0.88         0.87         0.87         9043
```

ROC AUC : 0.72

```
In [92]: from sklearn.metrics import roc_curve

fpr, tpr, thresholds = roc_curve(Ytest, Yprob)
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, label='ROC Curve', color='blue')
plt.title('ROC Curve for Decision Trees', fontsize=16)
plt.xlabel('False Positive Rate', fontsize=14)
plt.ylabel('True Positive Rate', fontsize=14)
plt.legend(loc='lower right')
plt.plot([0, 1], [0, 1], linestyle='--', color='red')
```

Out[92]: [<matplotlib.lines.Line2D at 0x137f93610>]



Model Evaluation:

- **Accuracy:** Higher than Logistic Regression
- **Precision & Recall:** Low grade, Big possibility of Overfitting
- **ROC AUC:** Normal, but not high

In General, This model is prone to overfitting since there is a possibility that Decision Tree learnt many nonlinear patterns

3.3 Random Forest

```
In [93]: pipe = imbPipeline(steps=[
            ('prep', preprocessor),
            ('smote', SMOTE(random_state=42)),
            ('RanFor', RandomForestClassifier(random_state=42))
        ])

pipe.fit(Xtrain, Ytrain)
Ypred = pipe.predict(Xtest)
Yprob = pipe.predict_proba(Xtest)[:, 1]
```

```
In [94]: from sklearn.metrics import roc_auc_score

print("          ===Classification Report===")
print(classification_report(Ytest, Ypred, target_names=['No', 'Yes']))
print(f"ROC AUC : {roc_auc_score(Ytest, Yprob):.2f}")
```

```
          ===Classification Report===
precision    recall  f1-score   support

       No       0.94      0.95      0.94       7985
       Yes       0.58      0.55      0.56       1058

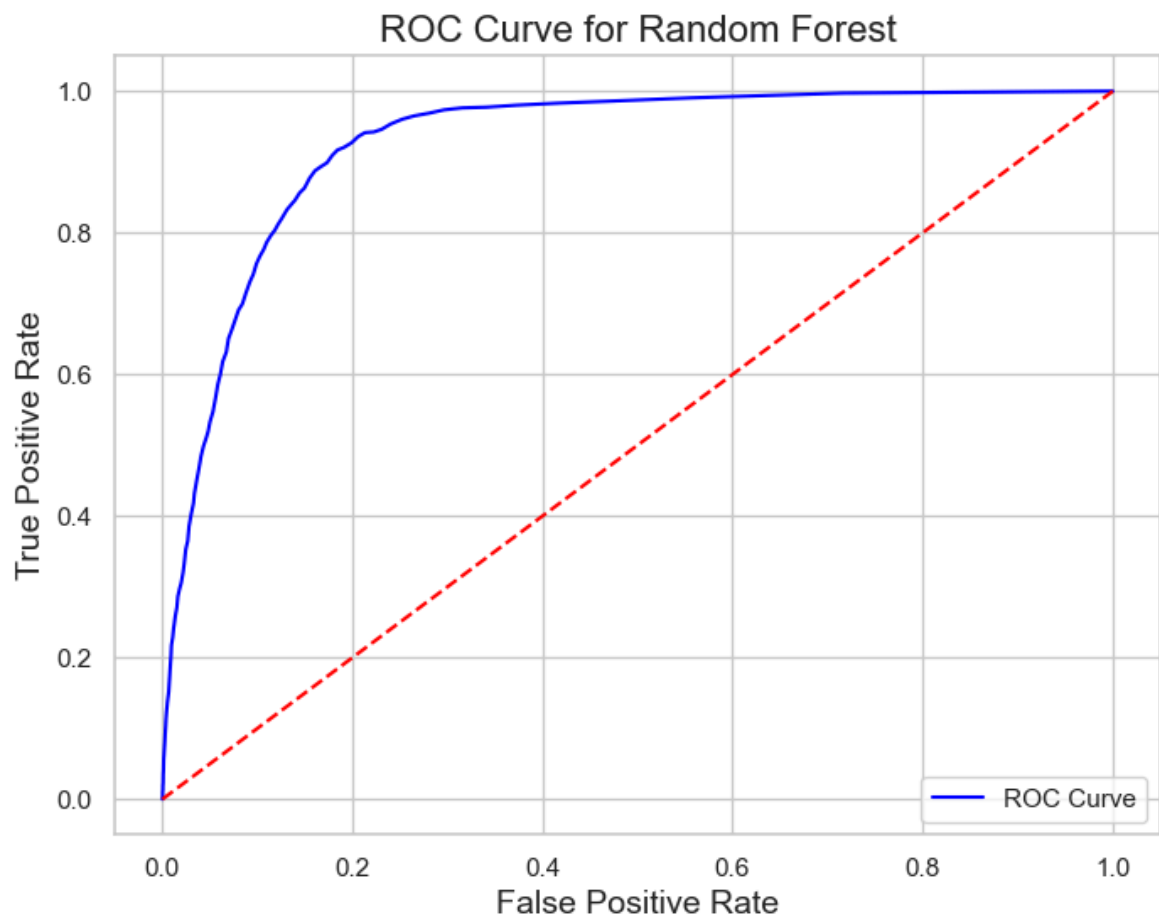
 accuracy          0.90       9043
 macro avg       0.76      0.75      0.75       9043
weighted avg       0.90      0.90      0.90       9043
```

ROC AUC : 0.93

```
In [95]: from sklearn.metrics import roc_curve

fpr, tpr, thresholds = roc_curve(Ytest, Yprob)
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, label='ROC Curve', color='blue')
plt.title('ROC Curve for Random Forest', fontsize=16)
plt.xlabel('False Positive Rate', fontsize=14)
plt.ylabel('True Positive Rate', fontsize=14)
plt.legend(loc='lower right')
plt.plot([0, 1], [0, 1], linestyle='--', color='red')
```

```
Out[95]: [<matplotlib.lines.Line2D at 0x169986710>]
```



Model Evaluation:

In General, This model did better than Decision Trees

3.4 XGBoost

```
In [96]: pipe = imbPipeline(steps=[
    ('prep', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('XGB', XGBClassifier(random_state=42, use_label_encoder=False, eval_m
    ])
```

```
pipe.fit(Xtrain, Ytrain)
Ypred = pipe.predict(Xtest)
Yprob = pipe.predict_proba(Xtest)[:, 1]
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-pac
kages/xgboost/training.py:183: UserWarning: [20:05:27] WARNING: /Users/run
ner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
In [97]: from sklearn.metrics import roc_auc_score

print("====Classification Report====")
print(classification_report(Ytest, Ypred, target_names=['No', 'Yes']))
print(f"ROC AUC : {roc_auc_score(Ytest, Yprob):.2f}")
```

```

===Classification Report===
precision    recall  f1-score   support

   No       0.94      0.95      0.95      7985
   Yes       0.61      0.57      0.59      1058

 accuracy          0.91      9043
 macro avg         0.78      0.76      0.77      9043
weighted avg         0.90      0.91      0.91      9043

```

ROC AUC : 0.93

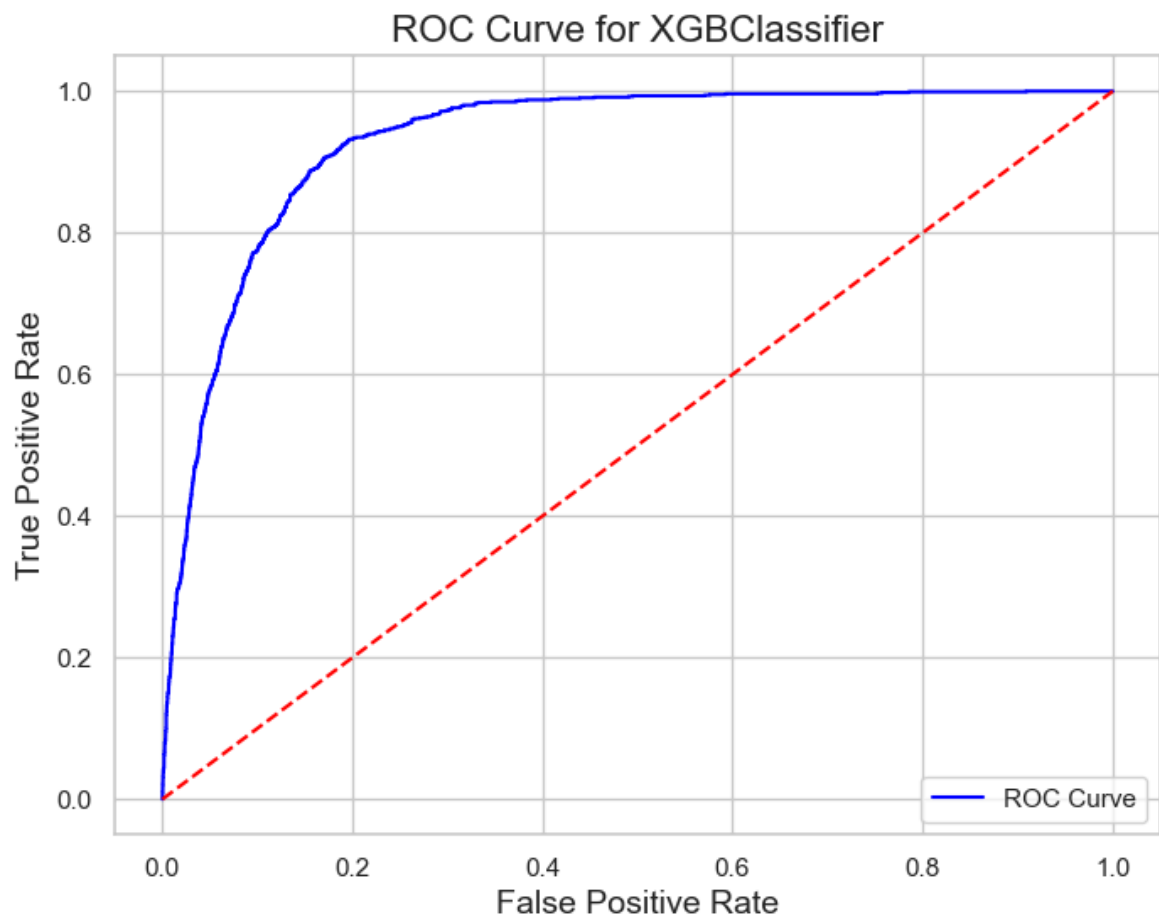
```

In [98]: from sklearn.metrics import roc_curve

fpr, tpr, thresholds = roc_curve(Ytest, Yprob)
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, label='ROC Curve', color='blue')
plt.title('ROC Curve for XGBClassifier', fontsize=16)
plt.xlabel('False Positive Rate', fontsize=14)
plt.ylabel('True Positive Rate', fontsize=14)
plt.legend(loc='lower right')
plt.plot([0, 1], [0, 1], linestyle='--', color='red')

```

Out[98]: [



Comparing with previous model, for now, results are the best

3.5 CatBoost

```

In [99]: pipe = imbPipeline(steps=[
          ('prep', preprocessor),

```

```

    ('smote', SMOTE(random_state=42)),
    ('CatBoost', CatBoostClassifier(verbose=0, random_state=42))
])

pipe.fit(Xtrain, Ytrain)
Ypred = pipe.predict(Xtest)
Yprob = pipe.predict_proba(Xtest)[:, 1]

```

```

In [100... from sklearn.metrics import roc_auc_score

print("====Classification Report====")
print(classification_report(Ytest, Ypred, target_names=['No', 'Yes']))
print(f"ROC AUC : {roc_auc_score(Ytest, Yprob):.2f}")

```

```

====Classification Report====
precision    recall  f1-score   support

      No      0.94      0.96      0.95       7985
      Yes      0.63      0.57      0.60       1058

 accuracy          0.91       9043
 macro avg       0.79      0.76      0.77       9043
weighted avg       0.91      0.91      0.91       9043

ROC AUC : 0.93

```

```

In [101... from sklearn.metrics import roc_curve

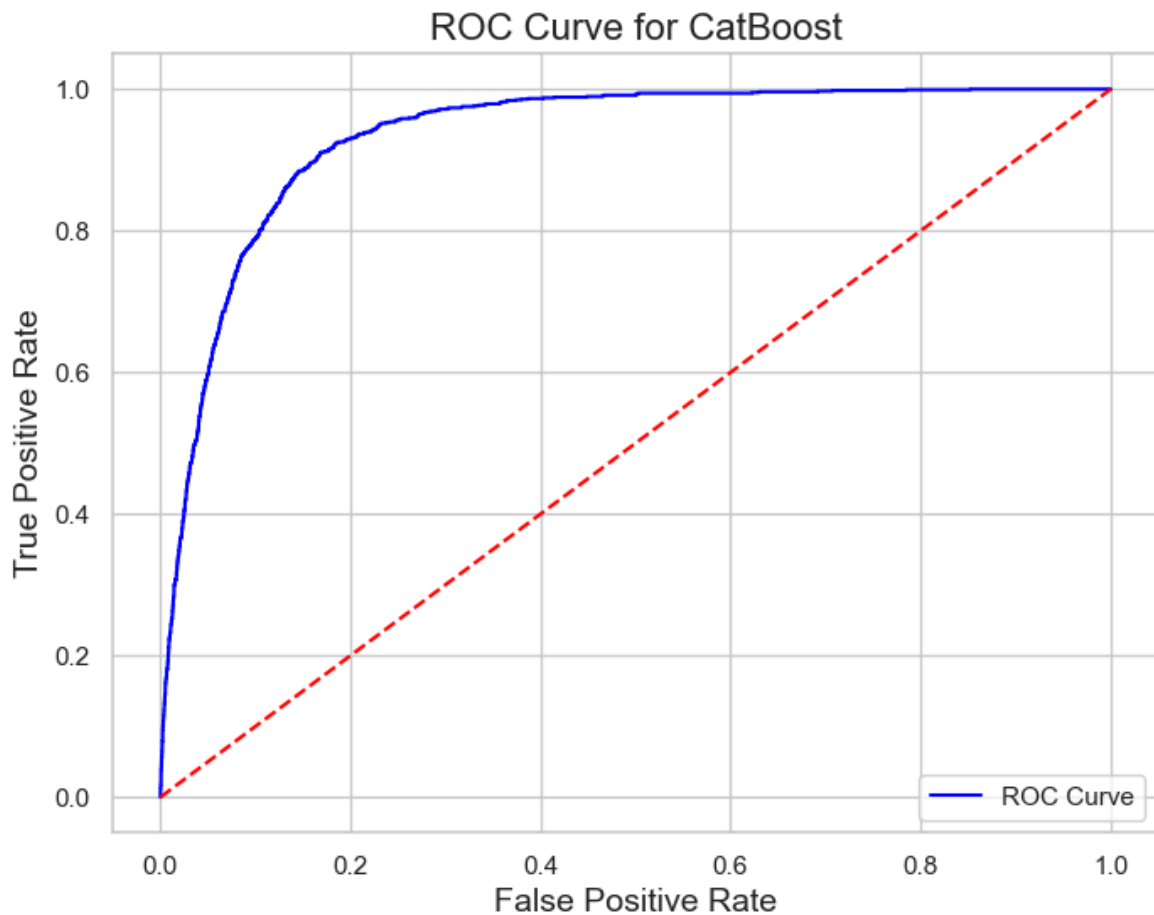
fpr, tpr, thresholds = roc_curve(Ytest, Yprob)
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, label='ROC Curve', color='blue')
plt.title('ROC Curve for CatBoost', fontsize=16)
plt.xlabel('False Positive Rate', fontsize=14)
plt.ylabel('True Positive Rate', fontsize=14)
plt.legend(loc='lower right')
plt.plot([0, 1], [0, 1], linestyle='--', color='red')

```

```

Out[101... [<matplotlib.lines.Line2D at 0x169af82d0>]

```



Good Results

4. Results comparison (Сравнение результатов)

Model Comparison

Model	Accuracy	Precision (Yes)	Recall (Yes)	F1-score (Yes)	ROC AUC
Logistic Regression	0.85	0.42	0.80	0.55	0.91
Decision Tree	0.87	0.44	0.54	0.48	0.72
Random Forest	0.90	0.58	0.55	0.56	0.93
XGBoost	0.91	0.68	0.57	0.59	0.93
CatBoost	0.91	0.63	0.57	0.60	0.93

Strengths and Weaknesses

1. Logistic Regression

- **Strengths:** High recall (0.80), simple, fast, easy to interpret.
- **Weaknesses:** Low precision (0.42), misses specificity, cannot model nonlinear relationships.

2. Decision Tree

- **Strengths:** Slightly better accuracy than LR, easy interpretability.
- **Weaknesses:** Lowest ROC AUC (0.72), overfits easily, weak generalization.

3. Random Forest

- **Strengths:** High accuracy (0.90), good precision (0.58), excellent ROC AUC (0.93), reduces overfitting compared to a single tree.
- **Weaknesses:** Less interpretable, longer training time than LR or single tree.

4. XGBoost

- **Strengths:** Highest precision (0.68), high accuracy (0.91), strong F1-score (0.59), excellent ROC AUC (0.93), handles complex data well.
- **Weaknesses:** Requires careful hyperparameter tuning, less interpretable, longer training time.

5. CatBoost

- **Strengths:** Highest F1-score (0.60), high precision (0.63), high accuracy (0.91), handles categorical features natively, excellent ROC AUC (0.93).
- **Weaknesses:** Longer training time, slightly less widespread in industry compared to XGBoost.

Best Model Selection

CatBoost is chosen as the best model for this task because:

- ✓ It achieved the **highest F1-score (0.60)**, indicating the best balance between precision and recall.
- ✓ Its precision (0.63) and recall (0.57) are both high, leading to reliable predictions.
- ✓ It handles categorical features efficiently without complex preprocessing.
- ✓ Its ROC AUC (0.93) is equal to XGBoost and Random Forest, showing strong class separation.

In summary: CatBoost provides **the most balanced and robust performance** for this classification task, making it the preferred choice.

5. Upgrading 2 best models

5.1 CatBoost

```
In [ ]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler, OneHotEncoder, MinMaxScaler
from sklearn.metrics import accuracy_score, precision_score, recall_score
from sklearn.metrics import classification_report, ConfusionMatrixDisplay
from sklearn.linear_model import LogisticRegression
```

```

from sklearn.compose import ColumnTransformer
from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.pipeline import Pipeline

from catboost import CatBoostClassifier
from xgboost import XGBClassifier

from imblearn.over_sampling import SMOTE
from imblearn.under_sampling import RandomUnderSampler
from imblearn.pipeline import Pipeline as imbPipeline

sns.set(style="whitegrid")

```

1. EDA

1.1 Data exploration

```
In [ ]: df = pd.read_csv("/Users/adilkhankaldybekov/Desktop/adilkhan/Machine Learning/credit_data.csv")
```

```
In [ ]: #Data Set observation
```

```
print(df.shape)
df.head()
```

(45211, 17)

	age	job	marital	education	default	balance	housing	loan	contact
0	58	management	married	tertiary	no	2143	yes	no	unknown
1	44	technician	single	secondary	no	29	yes	no	unknown
2	33	entrepreneur	married	secondary	no	2	yes	yes	unknown
3	47	blue-collar	married	unknown	no	1506	yes	no	unknown
4	33	unknown	single	unknown	no	1	no	no	unknown

```
In [ ]: df.info()
```



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 45211 entries, 0 to 45210
Data columns (total 17 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         45211 non-null  int64
1   job         45211 non-null  object
2   marital     45211 non-null  object
3   education   45211 non-null  object
4   default     45211 non-null  object
5   balance     45211 non-null  int64
6   housing     45211 non-null  object
7   loan        45211 non-null  object
8   contact     45211 non-null  object
9   day         45211 non-null  int64
10  month       45211 non-null  object
11  duration    45211 non-null  int64
12  campaign    45211 non-null  int64
13  pdays       45211 non-null  int64
14  previous    45211 non-null  int64
15  poutcome    45211 non-null  object
16  y           45211 non-null  object
dtypes: int64(7), object(10)
memory usage: 5.9+ MB
```

```
In [ ]: #Counting number of nan values for each feature
```

```
print(df.isna().sum())
```

```
age         0
job         0
marital     0
education   0
default     0
balance     0
housing     0
loan        0
contact     0
day         0
month       0
duration    0
campaign    0
pdays      0
previous    0
poutcome    0
y           0
dtype: int64
```

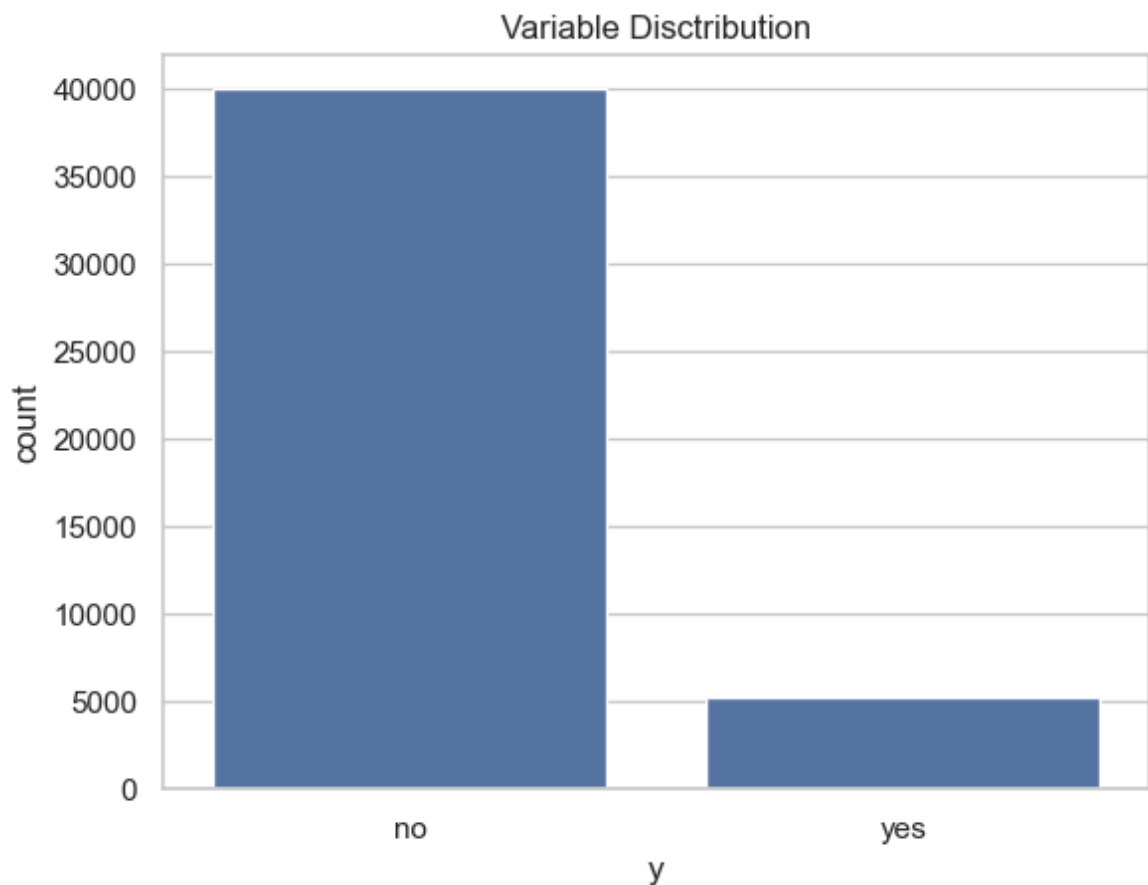
```
In [ ]: #Data observation for integer features
```

```
num_cols = df.select_dtypes(include=['int64','float64']).columns
df[num_cols].describe()
```

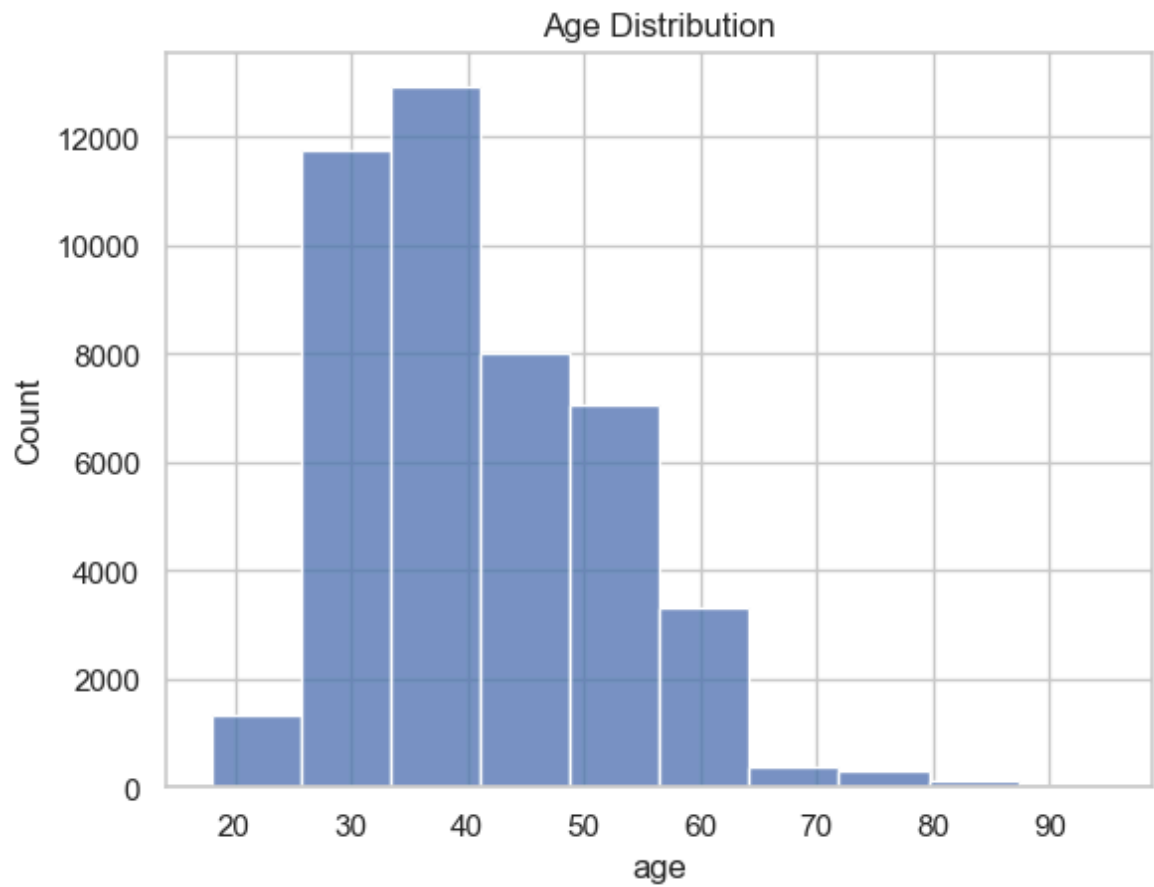
	age	balance	day	duration	campaign	
count	45211.000000	45211.000000	45211.000000	45211.000000	45211.000000	452
mean	40.936210	1362.272058	15.806419	258.163080	2.763841	
std	10.618762	3044.765829	8.322476	257.527812	3.098021	1
min	18.000000	-8019.000000	1.000000	0.000000	1.000000	
25%	33.000000	72.000000	8.000000	103.000000	1.000000	
50%	39.000000	448.000000	16.000000	180.000000	2.000000	
75%	48.000000	1428.000000	21.000000	319.000000	3.000000	
max	95.000000	102127.000000	31.000000	4918.000000	63.000000	8

1.2 Data visualization

```
In [ ]: #Distribution of all target values in dataset
sns.countplot(x='y', data=df)
plt.title('Variable Distribution')
plt.show()
```

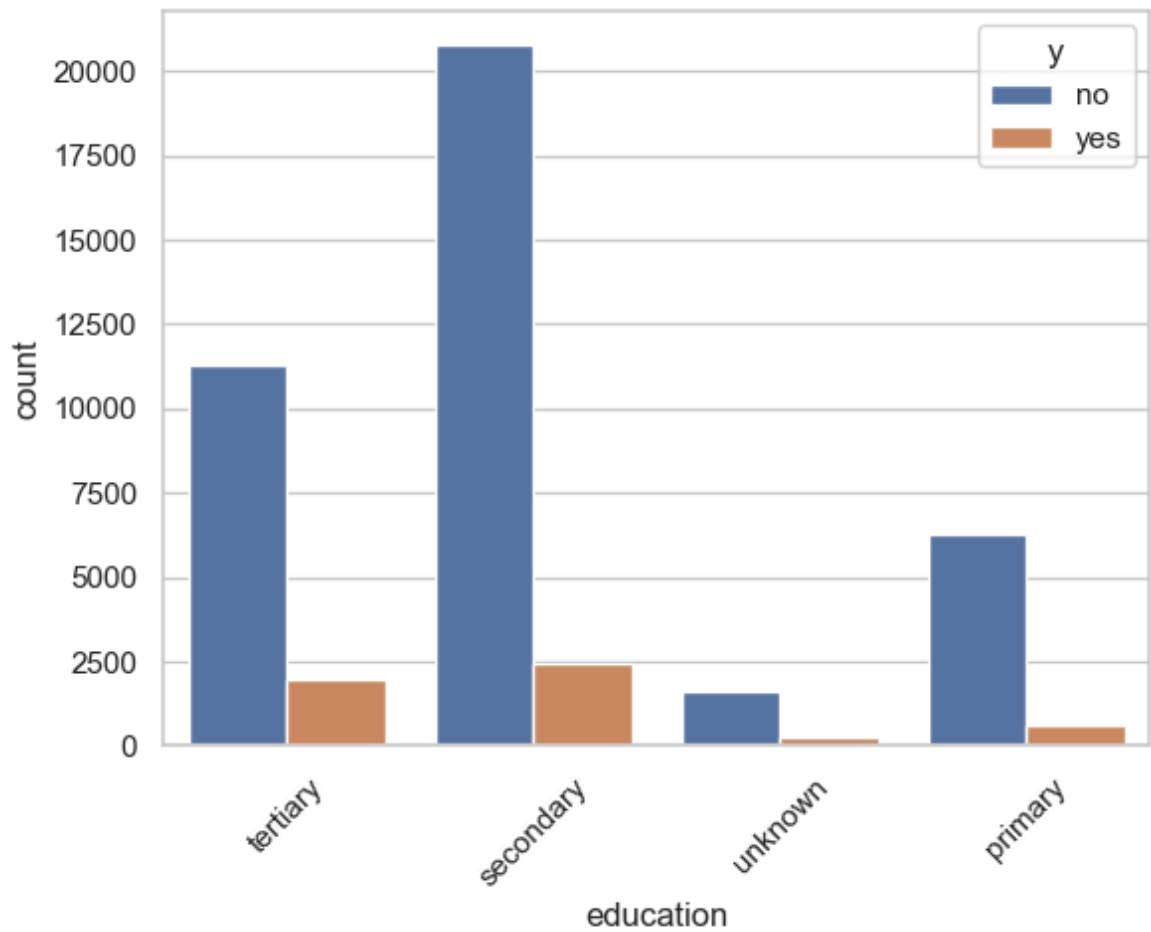


```
In [ ]: # Visualizing the distribution of 'age' column
sns.histplot(df['age'], bins=10)
plt.title('Age Distribution')
plt.show()
```

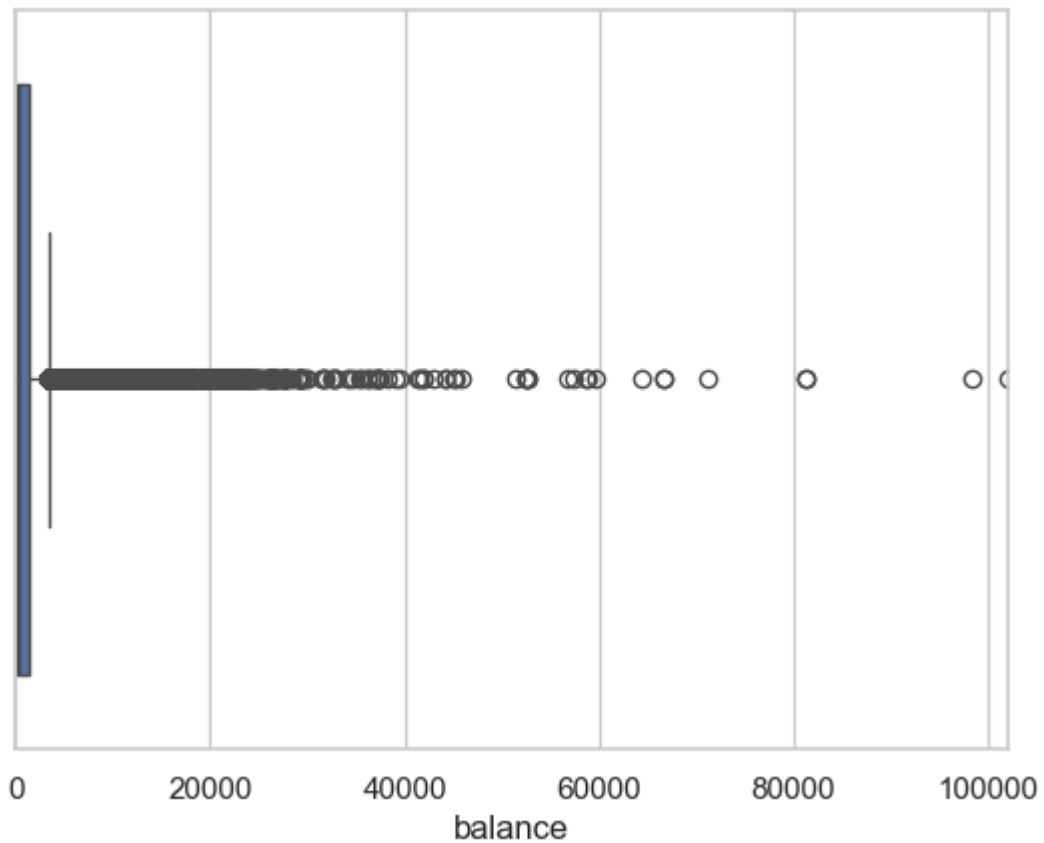


In []: *#Comparing of all education levels by target value*

```
sns.countplot(x='education', hue='y', data=df)
plt.xticks(rotation=45)
plt.show()
```



```
In [ ]: sns.boxplot(x=df['balance'])  
plt.xlim(0, df['balance'].max())  
plt.show()
```



1.3 Correlation Analysis

```
In [ ]: #looking at the relation between numerical features
correlation = df[num_cols].corr()
sns.heatmap(correlation, annot=True, fmt='.2f')
plt.title('Correlation Matrix')
plt.show()
```



2. Data processing/engineering

```
In [ ]: cat_cols = df.select_dtypes(include=['object']).drop(columns=['y']).columns
```

```
In [ ]: def create_prep():
    num_transformer = Pipeline(steps=[('scaler', StandardScaler())]) #for
    cat_transformer = Pipeline(steps=[('onehot', OneHotEncoder(handle_unknown='ignore'))])
    prep = ColumnTransformer(transformers=[('num', num_transformer, num_cols),
                                           ('cat', cat_transformer, cat_cols)])
    return prep

preprocessor = create_prep()
```

According to some observations we did data exploratory analysis and feature engineering engineering(Standardize data and Encoding)

2. Model Building (Построение моделей)

3.1 Logistic Regression

```
In [ ]: X = df.drop(columns='y', axis=1)
```

```
Y = (df['y'] == 'yes').astype(int)
#Y = df['y'].map({'yes': 1, 'no' : 0})
```

```
In [ ]: Xtrain, Xtest, Ytrain, Ytest = train_test_split( X, Y, test_size=0.2, ran
```

```
In [ ]: pipe = imbPipeline(steps=[
    ('prep', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('logreg', LogisticRegression(random_state=42, max_iter=1000))
])

pipe.fit(Xtrain, Ytrain)
Ypred = pipe.predict(Xtest)
Yprob = pipe.predict_proba(Xtest)[:, 1]

print("Distribution of values after SMOTE:")
Xsmote, Ysmote = pipe.named_steps['smote'].fit_resample(preprocessor.fit_
print(Ysmote.value_counts())
```

Distribution of values after SMOTE:

```
y
0    31937
1    31937
Name: count, dtype: int64
```

Evaluating

```
In [ ]: from sklearn.metrics import roc_auc_score

print("====Classification Report====")
print(classification_report(Ytest, Ypred, target_names=['No', 'Yes']))
print(f"ROC AUC : {roc_auc_score(Ytest, Yprob):.2f}")
```

```
====Classification Report====
precision    recall  f1-score   support

   No       0.97     0.85     0.91       7985
   Yes       0.42     0.80     0.55       1058

 accuracy          0.85       9043
 macro avg       0.69     0.83     0.73       9043
weighted avg       0.91     0.85     0.87       9043
```

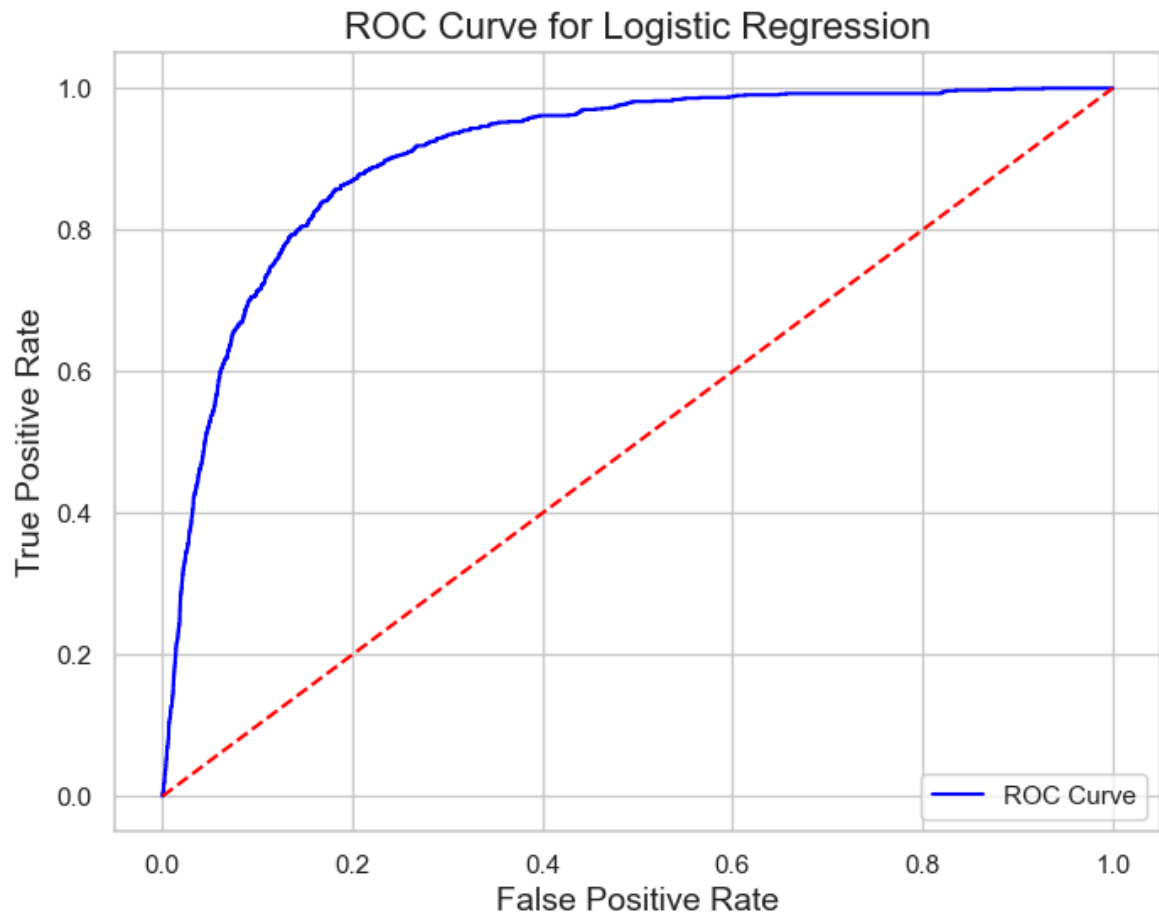
ROC AUC : 0.91

```
In [ ]: from sklearn.metrics import roc_curve

fpr, tpr, thresholds = roc_curve(Ytest, Yprob)
```

```
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, label='ROC Curve', color='blue')
plt.title('ROC Curve for Logistic Regression', fontsize=16)
plt.xlabel('False Positive Rate', fontsize=14)
plt.ylabel('True Positive Rate', fontsize=14)
plt.legend(loc='lower right')
plt.plot([0, 1], [0, 1], linestyle='--', color='red')
```

[<matplotlib.lines.Line2D at 0x1699f3c50>]



Model Evaluation:

- **Accuracy:** Good overall, predicts the majority class well
- **Precision:** Low - many false positives
- **Recall:** Good due to SMOTE
- **ROC AUC:** Normal, but not high

3.2 Decision Tree

```
In [ ]: pipe = imbPipeline(steps=[
    ('prep', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('DecTrees', DecisionTreeClassifier(random_state=42))
])

pipe.fit(Xtrain, Ytrain)
Ypred = pipe.predict(Xtest)
Yprob = pipe.predict_proba(Xtest)[:, 1]
```

```
In [ ]: from sklearn.metrics import roc_auc_score

print("====Classification Report====")
print(classification_report(Ytest, Ypred, target_names=['No', 'Yes']))
print(f"ROC AUC : {roc_auc_score(Ytest, Yprob):.2f}")
```

```
====Classification Report====
precision    recall  f1-score   support

     No       0.94      0.91      0.92      7985
     Yes       0.44      0.54      0.48      1058

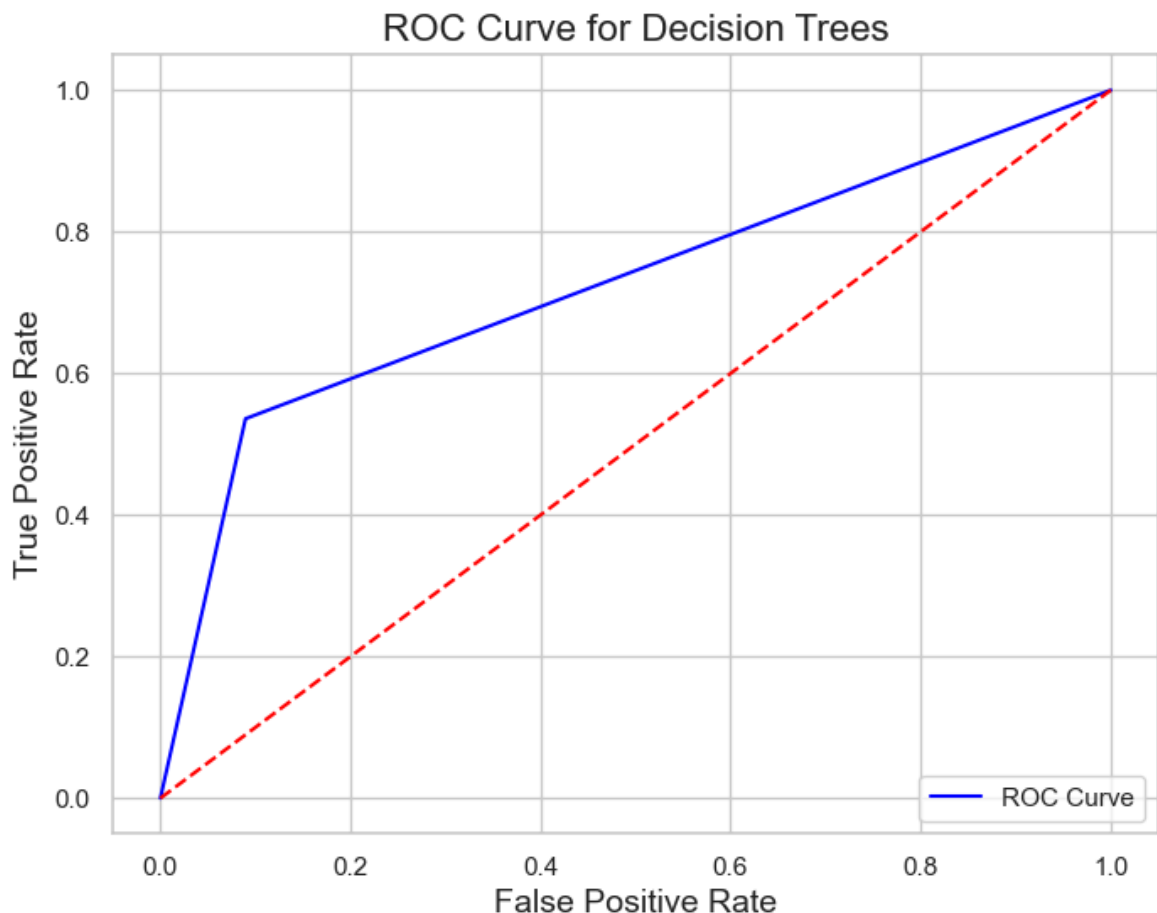
 accuracy          0.87      9043
 macro avg       0.69      0.72      0.70      9043
weighted avg       0.88      0.87      0.87      9043
```

ROC AUC : 0.72

```
In [ ]: from sklearn.metrics import roc_curve

fpr, tpr, thresholds = roc_curve(Ytest, Yprob)
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, label='ROC Curve', color='blue')
plt.title('ROC Curve for Decision Trees', fontsize=16)
plt.xlabel('False Positive Rate', fontsize=14)
plt.ylabel('True Positive Rate', fontsize=14)
plt.legend(loc='lower right')
plt.plot([0, 1], [0, 1], linestyle='--', color='red')
```

[<matplotlib.lines.Line2D at 0x137f93610>]



Model Evaluation:

- **Accuracy:** Higher than Logistic Regression
- **Precision & Recall:** Low grade, Big possibility of Overfitting
- **ROC AUC:** Normal, but not high

In General, This model is prone to overfitting since there is a possibility that Decision Tree learnt many nonlinear patterns

3.3 Random Forest

```
In [ ]: pipe = imbPipeline(steps=[
    ('prep', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('RanFor', RandomForestClassifier(random_state=42))
])

pipe.fit(Xtrain, Ytrain)
Ypred = pipe.predict(Xtest)
Yprob = pipe.predict_proba(Xtest)[:, 1]
```

```
In [ ]: from sklearn.metrics import roc_auc_score

print("          ===Classification Report===")
print(classification_report(Ytest, Ypred, target_names=['No', 'Yes']))
print(f"ROC AUC : {roc_auc_score(Ytest, Yprob):.2f}")
```

```
          ===Classification Report===
precision    recall  f1-score   support

   No         0.94     0.95     0.94       7985
   Yes         0.58     0.55     0.56       1058

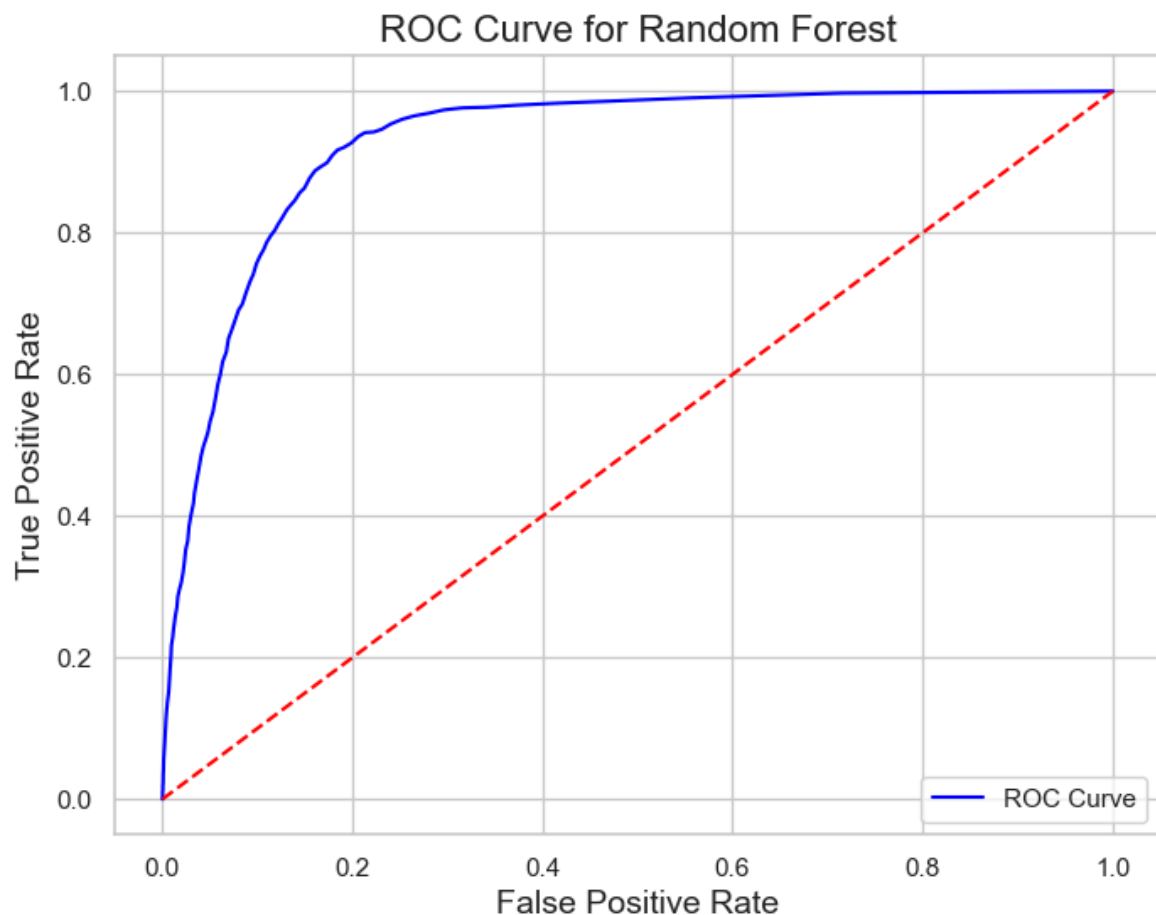
 accuracy          0.90       9043
 macro avg         0.76     0.75     0.75       9043
weighted avg         0.90     0.90     0.90       9043
```

ROC AUC : 0.93

```
In [ ]: from sklearn.metrics import roc_curve

fpr, tpr, thresholds = roc_curve(Ytest, Yprob)
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, label='ROC Curve', color='blue')
plt.title('ROC Curve for Random Forest', fontsize=16)
plt.xlabel('False Positive Rate', fontsize=14)
plt.ylabel('True Positive Rate', fontsize=14)
plt.legend(loc='lower right')
plt.plot([0, 1], [0, 1], linestyle='--', color='red')
```

[<matplotlib.lines.Line2D at 0x169986710>]



Model Evaluation:

In General, This model did better than Decision Trees

3.4 XGBoost

```
In [ ]: pipe = imbPipeline(steps=[
    ('prep', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('XGB', XGBClassifier(random_state=42, use_label_encoder=False, eval_m
    ])
```

```
pipe.fit(Xtrain, Ytrain)
Ypred = pipe.predict(Xtest)
Yprob = pipe.predict_proba(Xtest)[:, 1]
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-pac
kages/xgboost/training.py:183: UserWarning: [20:05:27] WARNING: /Users/run
ner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
In [ ]: from sklearn.metrics import roc_auc_score

print("====Classification Report====")
print(classification_report(Ytest, Ypred, target_names=['No', 'Yes']))
print(f"ROC AUC : {roc_auc_score(Ytest, Yprob):.2f}")
```

```

===Classification Report===
precision    recall  f1-score   support

   No       0.94      0.95      0.95      7985
   Yes       0.61      0.57      0.59      1058

 accuracy          0.91      9043
 macro avg         0.78      0.76      0.77      9043
weighted avg         0.90      0.91      0.91      9043

```

ROC AUC : 0.93

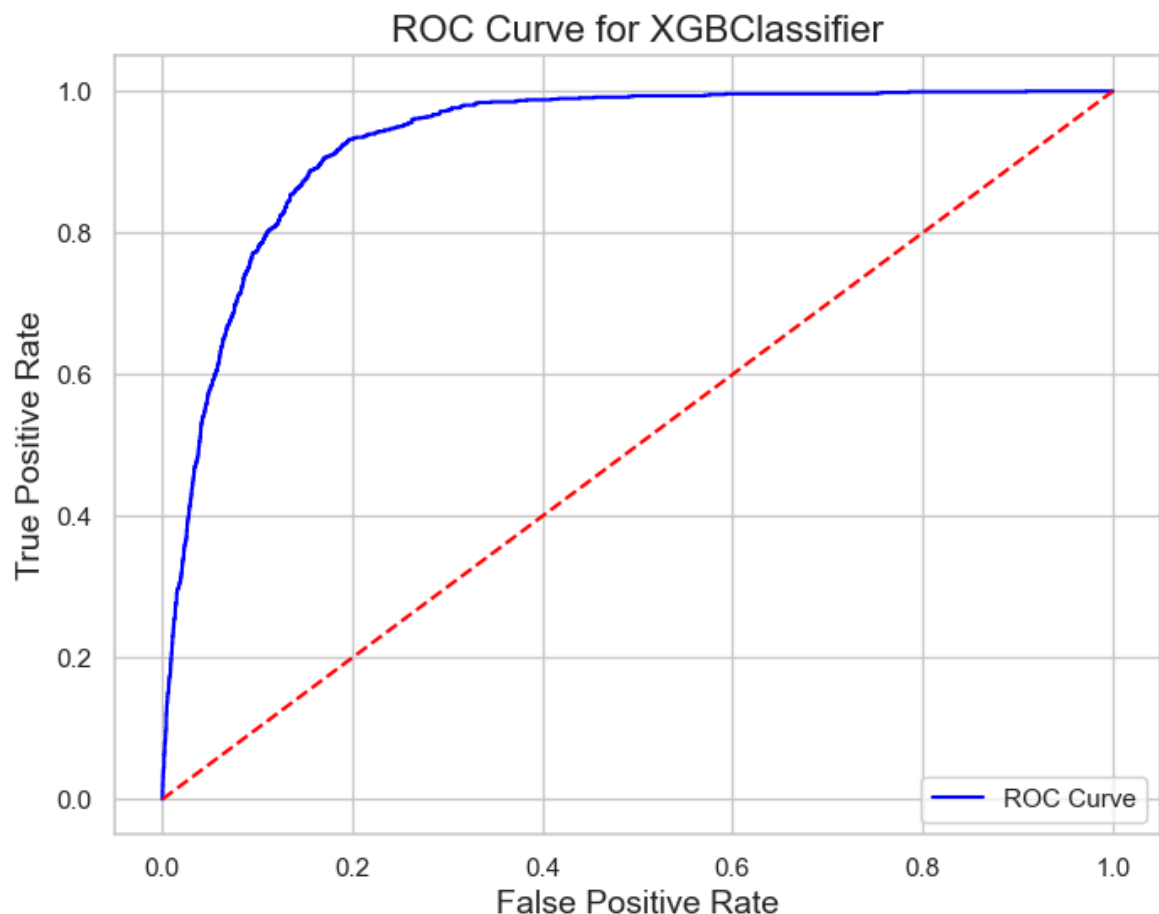
```

In [ ]: from sklearn.metrics import roc_curve

fpr, tpr, thresholds = roc_curve(Ytest, Yprob)
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, label='ROC Curve', color='blue')
plt.title('ROC Curve for XGBClassifier', fontsize=16)
plt.xlabel('False Positive Rate', fontsize=14)
plt.ylabel('True Positive Rate', fontsize=14)
plt.legend(loc='lower right')
plt.plot([0, 1], [0, 1], linestyle='--', color='red')

```

[matplotlib.lines.Line2D at 0x169a75590>]



Comparing with previous model, for now, results are the best

3.5 CatBoost

```

In [ ]: pipe = imbPipeline(steps=[
    ('prep', preprocessor),
    ('smote', SMOTE(random_state=42)),

```

```

    ('CatBoost', CatBoostClassifier(verbose=0, random_state=42))
])

pipe.fit(Xtrain, Ytrain)
Ypred = pipe.predict(Xtest)
Yprob = pipe.predict_proba(Xtest)[:, 1]

```

```

In [ ]: from sklearn.metrics import roc_auc_score

print("====Classification Report====")
print(classification_report(Ytest, Ypred, target_names=['No', 'Yes']))
print(f"ROC AUC : {roc_auc_score(Ytest, Yprob):.2f}")

```

```

====Classification Report====
precision    recall  f1-score   support

     No       0.94      0.96      0.95      7985
     Yes       0.63      0.57      0.60      1058

 accuracy          0.91      9043
 macro avg       0.79      0.76      0.77      9043
weighted avg       0.91      0.91      0.91      9043

```

ROC AUC : 0.93

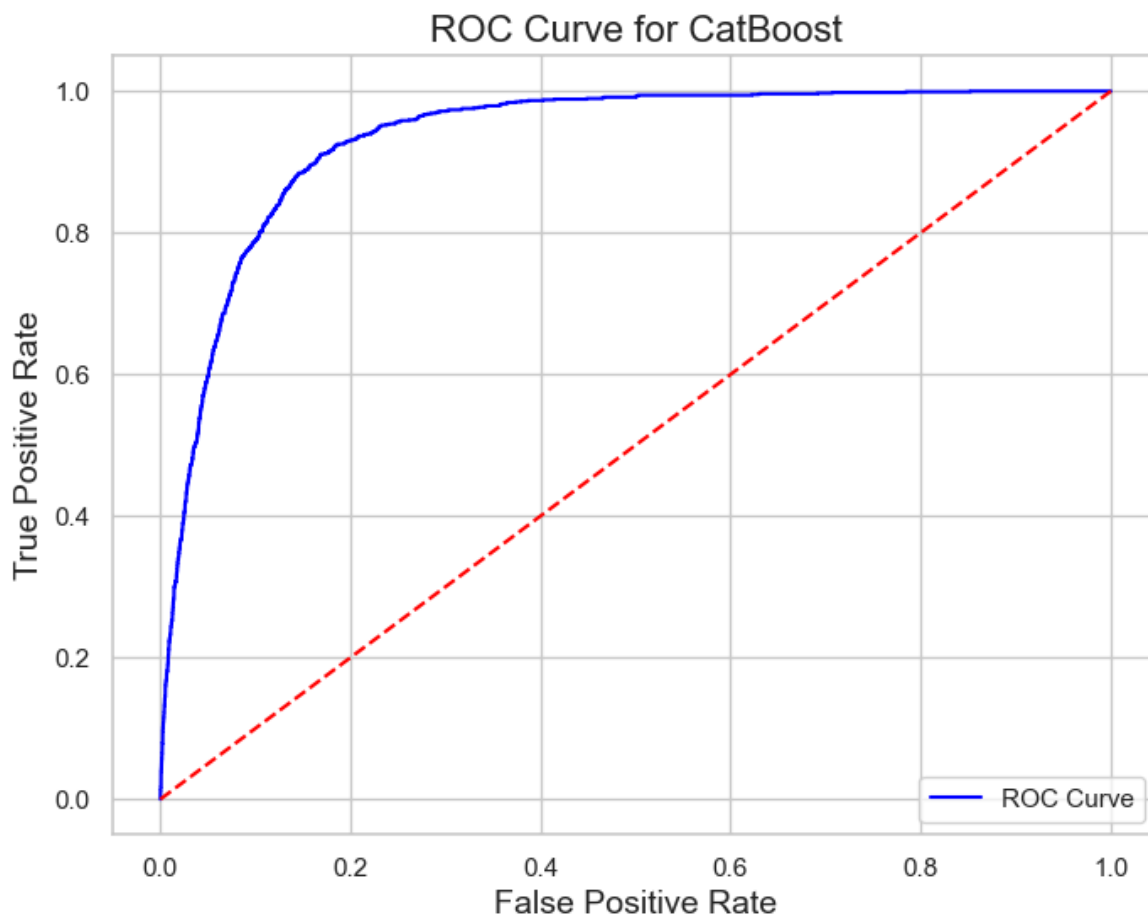
```

In [ ]: from sklearn.metrics import roc_curve

fpr, tpr, thresholds = roc_curve(Ytest, Yprob)
plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, label='ROC Curve', color='blue')
plt.title('ROC Curve for CatBoost', fontsize=16)
plt.xlabel('False Positive Rate', fontsize=14)
plt.ylabel('True Positive Rate', fontsize=14)
plt.legend(loc='lower right')
plt.plot([0, 1], [0, 1], linestyle='--', color='red')

```

[<matplotlib.lines.Line2D at 0x169af82d0>]



Good Results

4. Results comparison (Сравнение результатов)

Model Comparison

Model	Accuracy	Precision (Yes)	Recall (Yes)	F1-score (Yes)	ROC AUC
Logistic Regression	0.85	0.42	0.80	0.55	0.91
Decision Tree	0.87	0.44	0.54	0.48	0.72
Random Forest	0.90	0.58	0.55	0.56	0.93
XGBoost	0.91	0.68	0.57	0.59	0.93
CatBoost	0.91	0.63	0.57	0.60	0.93

Strengths and Weaknesses

1. Logistic Regression

- **Strengths:** High recall (0.80), simple, fast, easy to interpret.
- **Weaknesses:** Low precision (0.42), misses specificity, cannot model nonlinear relationships.

2. Decision Tree

- **Strengths:** Slightly better accuracy than LR, easy interpretability.
- **Weaknesses:** Lowest ROC AUC (0.72), overfits easily, weak generalization.

3. Random Forest

- **Strengths:** High accuracy (0.90), good precision (0.58), excellent ROC AUC (0.93), reduces overfitting compared to a single tree.
- **Weaknesses:** Less interpretable, longer training time than LR or single tree.

4. XGBoost

- **Strengths:** Highest precision (0.68), high accuracy (0.91), strong F1-score (0.59), excellent ROC AUC (0.93), handles complex data well.
- **Weaknesses:** Requires careful hyperparameter tuning, less interpretable, longer training time.

5. CatBoost

- **Strengths:** Highest F1-score (0.60), high precision (0.63), high accuracy (0.91), handles categorical features natively, excellent ROC AUC (0.93).
- **Weaknesses:** Longer training time, slightly less widespread in industry compared to XGBoost.

Best Model Selection

CatBoost is chosen as the best model for this task because:

- ✓ It achieved the **highest F1-score (0.60)**, indicating the best balance between precision and recall.
- ✓ Its precision (0.63) and recall (0.57) are both high, leading to reliable predictions.
- ✓ It handles categorical features efficiently without complex preprocessing.
- ✓ Its ROC AUC (0.93) is equal to XGBoost and Random Forest, showing strong class separation.

In summary: CatBoost provides **the most balanced and robust performance** for this classification task, making it the preferred choice.

5. Upgrading 2 best models

5.1 CatBoost

```
In [ ]: from sklearn.model_selection import cross_val_score
from imblearn.pipeline import Pipeline as ImbPipeline
from imblearn.over_sampling import SMOTE
from catboost import CatBoostClassifier
import numpy as np

pipe_cb = ImbPipeline(steps=[
    ('prep',      preprocessor),
    ('smote',     SMOTE(random_state=42)),
    ('clf',       CatBoostClassifier(verbose=0, random_state=42))
])
```

```

])

cv_scores_cb = cross_val_score(pipe_cb, Xtrain, Ytrain,
                                cv=5, scoring='f1', n_jobs=-1)
print(f"CatBoost 5-Fold CV F1-score: {cv_scores_cb.mean():.3f} ± {cv_scores_cb.std():.3f}")

```

CatBoost 5-Fold CV F1-score: 0.596 ± 0.012

```

In [ ]: from sklearn.model_selection import GridSearchCV

param_grid_cb = {
    'clf__depth':      [3, 5, 7],
    'clf__learning_rate': [0.01, 0.1, 0.2],
    'clf__iterations':  [100, 200]
}

grid_cb = GridSearchCV(pipe_cb,
                        param_grid_cb,
                        cv=5,
                        scoring='f1',
                        n_jobs=-1,
                        verbose=1)

grid_cb.fit(Xtrain, Ytrain)

print("Best CatBoost Parameters:", grid_cb.best_params_)
print(f"Best CatBoost CV F1-score: {grid_cb.best_score_: .3f}")

```

Fitting 5 folds for each of 18 candidates, totalling 90 fits

Best CatBoost Parameters: {'clf__depth': 5, 'clf__iterations': 100, 'clf__learning_rate': 0.1}

Best CatBoost CV F1-score: 0.609

```

In [ ]: final_cb = ImbPipeline(steps=[
    ('prep',  preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('clf',   CatBoostClassifier(
        verbose=0,
        random_state=42,
        depth=5,
        iterations=100,
        learning_rate=0.1
    ))
])

final_cb.fit(Xtrain, Ytrain)

y_pred_cb = final_cb.predict(Xtest)
y_proba_cb = final_cb.predict_proba(Xtest)[:, 1]

```

```
Exception ignored in: <function ResourceTracker.__del__ at 0x104c65440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x104b91440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x110159440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x106a15440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x105f75440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x1064a1440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x1044bd440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
```



```

File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x102f29440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes

```

```

In [ ]: print("      === Final CatBoost Classification Report ===")
print(classification_report(Ytest, y_pred_cb))
print(f"ROC AUC : {roc_auc_score(Ytest, y_proba_cb):.2f}")

fpr, tpr, _ = roc_curve(Ytest, y_proba_cb)
plt.figure(figsize=(7,5))
plt.plot(fpr, tpr, label=f'CatBoost (AUC = {roc_auc_score(Ytest, y_proba_
plt.plot([0,1],[0,1], '--', color='red')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve – Final CatBoost')
plt.legend(loc='lower right')
plt.grid(True)
plt.show()

```

```

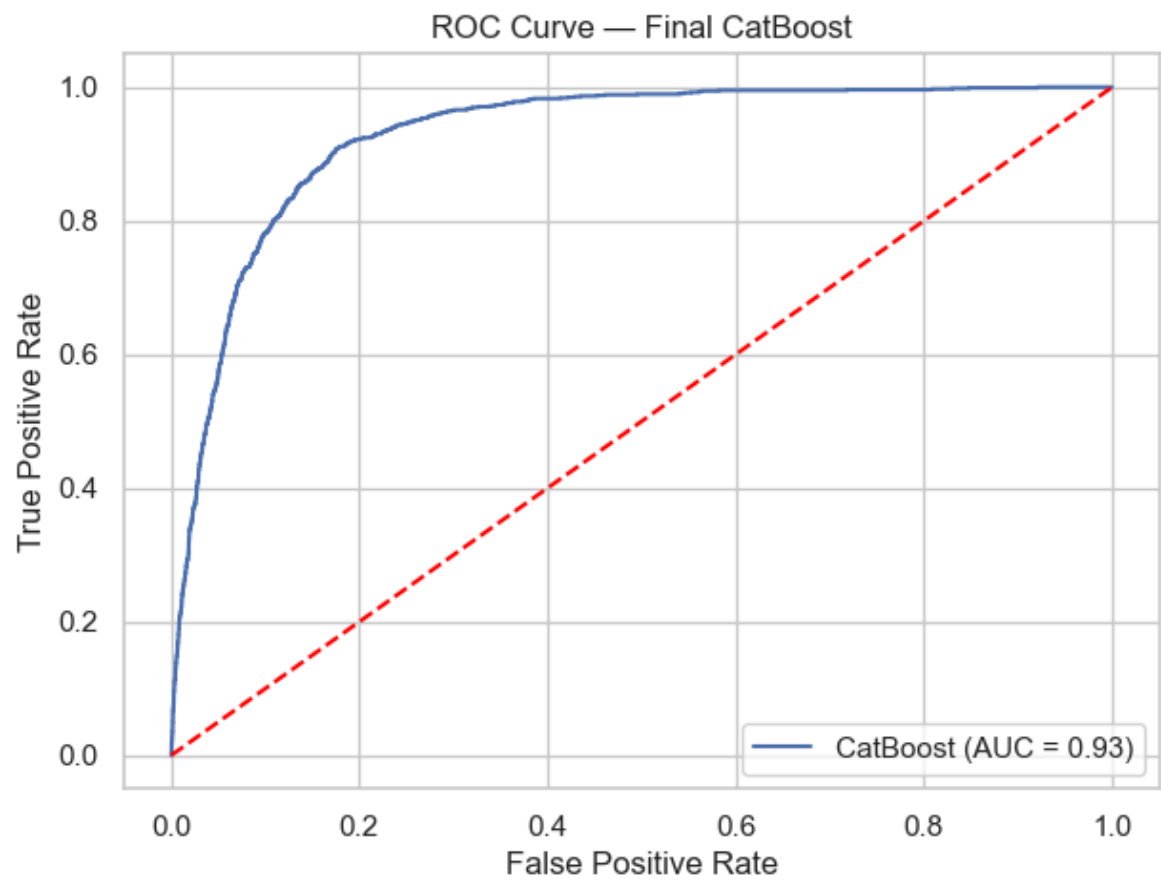
      === Final CatBoost Classification Report ===
              precision    recall  f1-score   support

         0       0.96      0.93      0.94       7985
         1       0.57      0.69      0.63       1058

 accuracy          0.90       0.90       0.90       9043
 macro avg       0.76      0.81      0.79       9043
weighted avg       0.91      0.90      0.91       9043

```

ROC AUC : 0.93



```
Exception ignored in: <function ResourceTracker.__del__ at 0x103091440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x106359440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x102bad440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x104359440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x104ff5440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x102515440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x1050e5440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
```

```

File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x102db5440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes

```

5.2 XGBClassifier

```

In [ ]: pipe_xgb = ImbPipeline(steps=[
    ('prep', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('clf', XGBClassifier(
        use_label_encoder=False,
        eval_metric='logloss',
        random_state=42
    ))
])

scores = cross_val_score(
    pipe_xgb,
    Xtrain, Ytrain,
    cv=5,
    scoring='f1',
    n_jobs=-1
)
print(f"XGBoost 5-Fold CV F1-score: {scores.mean():.3f} ± {scores.std():.3f}")

```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
XGBoost 5-Fold CV F1-score: 0.582 ± 0.013
```

```
In [ ]: param_grid_xgb = {
        'clf__max_depth':    [3, 5, 7],
        'clf__learning_rate': [0.01, 0.1, 0.2],
        'clf__n_estimators':  [100, 200],
        'clf__subsample':     [0.8, 1.0],
        'clf__colsample_bytree': [0.8, 1.0]
    }

    grid_xgb = GridSearchCV(
        pipe_xgb,
        param_grid=param_grid_xgb,
        cv=5,
        scoring='f1',
        n_jobs=-1,
        verbose=1
    )
    grid_xgb.fit(Xtrain, Ytrain)

    print("Best XGBoost Parameters:", grid_xgb.best_params_)
    print(f"Best CV F1-score after tuning: {grid_xgb.best_score_:.3f}")
```

Fitting 5 folds for each of 72 candidates, totalling 360 fits

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:53] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:53] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:53] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:53] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:53] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:54] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:54] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:54] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:54] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:57] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:00] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:03] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:03] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:03] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:03] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:04] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:04] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:09] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:09] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:09] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:09] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:09] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:09] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:09] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:11] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:11] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:13] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:13] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:15] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:16] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:16] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:16] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:19] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:19] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:19] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:20] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:20] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:20] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:20] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:20] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:20] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:21] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:21] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:21] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:21] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:21] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:21] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:21] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:22] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:22] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:22] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:23] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:23] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:23] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:23] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:23] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:23] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:25] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:25] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:25] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:25] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:27] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:27] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:27] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:28] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:28] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:29] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:29] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:29] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:29] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:29] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:30] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:30] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:30] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:31] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:31] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:31] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:31] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:33] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:34] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:34] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:34] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:35] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:35] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:35] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:35] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:35] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:35] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:36] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:36] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:36] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:36] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:36] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:36] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:37] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:37] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:37] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:37] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:38] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:38] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:38] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:38] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:39] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:39] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:39] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:39] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:40] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:40] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:40] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:40] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:41] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:41] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:41] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:41] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:42] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:42] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:42] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:42] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:44] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:44] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:45] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:45] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:45] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:45] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:45] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:46] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:48] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:48] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:48] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:00] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:00] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:02] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:03] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:03] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:04] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:04] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:04] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:04] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:04] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:13] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:15] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:22] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:23] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:25] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:25] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:29] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:30] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:31] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:33] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:39] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:39] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:40] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:41] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:43] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:43] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:44] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:44] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:44] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:45] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:46] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:46] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:47] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:47] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:47] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:48] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:53] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:54] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:54] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:55] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:57] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:57] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:57] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:00] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:00] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:13] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:13] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:13] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:13] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:19] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:28] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
Best XGBoost Parameters: {'clf__colsample_bytree': 0.8, 'clf__learning_rate': 0.1, 'clf__max_depth': 7, 'clf__n_estimators': 100, 'clf__subsample': 0.8}
```

```
Best CV F1-score after tuning: 0.610
```

```
In [ ]: final_xgb = ImbPipeline(steps=[
    ('prep', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('clf', XGBClassifier(
        use_label_encoder=False,
        eval_metric='logloss',
        learning_rate=0.1,
        max_depth=7,
        n_estimators=100,
        subsample=0.8,
        colsample_bytree=0.8,
        random_state=42
    ))
])

final_xgb.fit(Xtrain, Ytrain)

y_pred = final_xgb.predict(Xtest)
y_proba = final_xgb.predict_proba(Xtest)[:, 1]
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:19:39] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
In [ ]: print("          === Final XGBoost Classification Report ===")
print(classification_report(Ytest, y_pred, target_names=['No', 'Yes']))
print(f"ROC AUC : {roc_auc_score(Ytest, y_proba):.2f}")

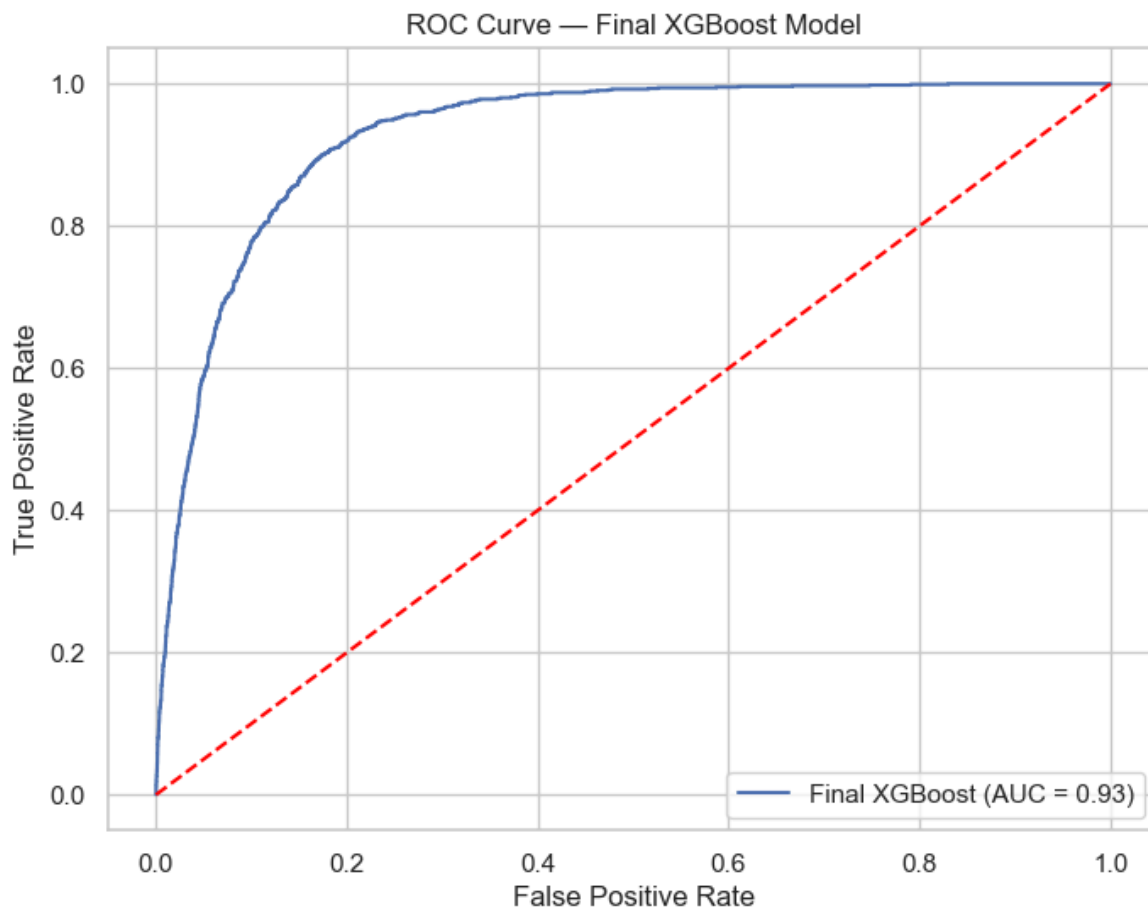
fpr, tpr, _ = roc_curve(Ytest, y_proba)
plt.figure(figsize=(8,6))
plt.plot(fpr, tpr, label=f'Final XGBoost (AUC = {roc_auc_score(Ytest, y_p
plt.plot([0,1],[0,1], '--', color='red')
plt.title('ROC Curve - Final XGBoost Model')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend(loc='lower right')
plt.grid(True)
plt.show()
```

```
=== Final XGBoost Classification Report ===
      precision    recall  f1-score   support

   No       0.96       0.93       0.94       7985
   Yes       0.57       0.67       0.62       1058

 accuracy          0.90       9043
  macro avg       0.76       0.80       0.78       9043
 weighted avg     0.91       0.90       0.91       9043
```

ROC AUC : 0.93



Comparison

Metric	CatBoost	XGBoost
Accuracy	0.90	0.90
Precision (Yes)	0.57	0.57
Recall (Yes)	0.69	0.67
F1-score (Yes)	0.63	0.62
ROC AUC	0.928	0.929
5-Fold CV F1	0.596 ± 0.012	0.582 ± 0.013

Overall, both tuned CatBoost and XGBoost deliver very strong performance on this imbalanced classification task, achieving nearly identical accuracy (90%) and AUC (≈ 0.93). CatBoost pulls ahead with a slightly better F1-score (0.63 vs. 0.62) and higher recall (0.69 vs. 0.67) on the positive ("yes") class, suggesting it strikes a

marginally better balance between catching true positives and avoiding false alarms. XGBoost, in turn, edges out by a hair on AUC and often trains a bit faster, making it attractive when throughput is critical. If your highest priority is maximizing the F1-score on the minority class and you value native handling of categorical features, CatBoost is the way to go; if you need the smallest possible latency or align with an existing XGBoost-based pipeline, XGBoost remains an excellent choice.

```
In [102... from sklearn.model_selection import cross_val_score
from imblearn.pipeline import Pipeline as ImbPipeline
from imblearn.over_sampling import SMOTE
from catboost import CatBoostClassifier
import numpy as np

pipe_cb = ImbPipeline(steps=[
    ('prep',      preprocessor),
    ('smote',     SMOTE(random_state=42)),
    ('clf',       CatBoostClassifier(verbose=0, random_state=42))
])

cv_scores_cb = cross_val_score(pipe_cb, Xtrain, Ytrain,
                               cv=5, scoring='f1', n_jobs=-1)
print(f"CatBoost 5-Fold CV F1-score: {cv_scores_cb.mean():.3f} ± {cv_scor

CatBoost 5-Fold CV F1-score: 0.596 ± 0.012
```

```
In [103... from sklearn.model_selection import GridSearchCV

param_grid_cb = {
    'clf__depth':      [3, 5, 7],
    'clf__learning_rate': [0.01, 0.1, 0.2],
    'clf__iterations': [100, 200]
}

grid_cb = GridSearchCV(pipe_cb,
                       param_grid_cb,
                       cv=5,
                       scoring='f1',
                       n_jobs=-1,
                       verbose=1)
grid_cb.fit(Xtrain, Ytrain)

print("Best CatBoost Parameters:", grid_cb.best_params_)
print(f"Best CatBoost CV F1-score: {grid_cb.best_score_:.3f}")
```

Fitting 5 folds for each of 18 candidates, totalling 90 fits
 Best CatBoost Parameters: {'clf__depth': 5, 'clf__iterations': 100, 'clf__learning_rate': 0.1}
 Best CatBoost CV F1-score: 0.609

```
In [109... final_cb = ImbPipeline(steps=[
    ('prep',      preprocessor),
    ('smote',     SMOTE(random_state=42)),
    ('clf',       CatBoostClassifier(
        verbose=0,
        random_state=42,
        depth=5,
        iterations=100,
        learning_rate=0.1
```

```
        ))  
    ])  
  
    final_cb.fit(Xtrain, Ytrain)  
  
    y_pred_cb = final_cb.predict(Xtest)  
    y_proba_cb = final_cb.predict_proba(Xtest)[:, 1]
```

```
Exception ignored in: <function ResourceTracker.__del__ at 0x104c65440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x104b91440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x110159440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x106a15440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x105f75440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x1064a1440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x1044bd440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
```

```

File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x102f29440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 82, in __del__
    File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 91, in _stop
    File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes

```

```

In [118... print("      === Final CatBoost Classification Report ===")
print(classification_report(Ytest, y_pred_cb))
print(f"ROC AUC : {roc_auc_score(Ytest, y_proba_cb):.2f}")

fpr, tpr, _ = roc_curve(Ytest, y_proba_cb)
plt.figure(figsize=(7,5))
plt.plot(fpr, tpr, label=f'CatBoost (AUC = {roc_auc_score(Ytest, y_proba_
plt.plot([0,1],[0,1], '--', color='red')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve – Final CatBoost')
plt.legend(loc='lower right')
plt.grid(True)
plt.show()

```

```

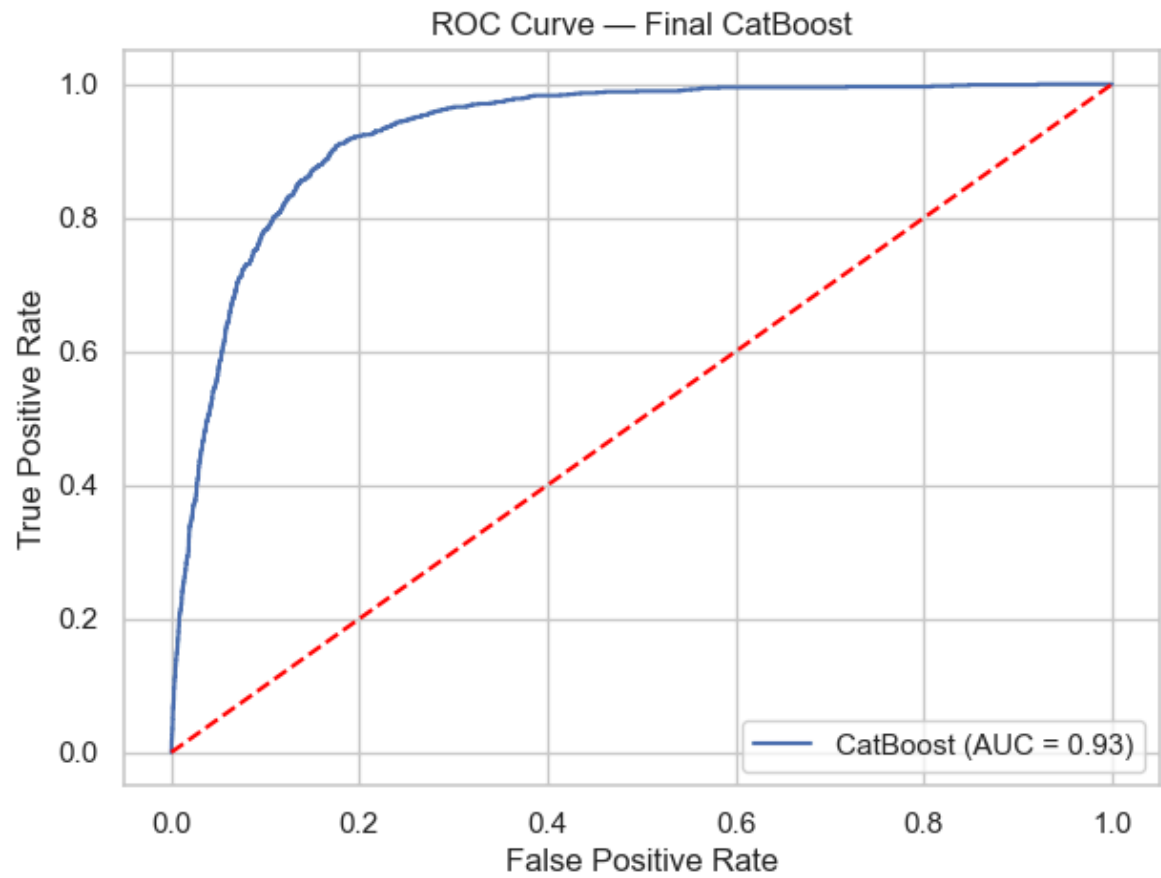
      === Final CatBoost Classification Report ===
              precision    recall  f1-score   support

         0       0.96      0.93      0.94       7985
         1       0.57      0.69      0.63       1058

 accuracy          0.90       9043
 macro avg       0.76      0.81      0.79       9043
 weighted avg    0.91      0.90      0.91       9043

```

ROC AUC : 0.93



```
Exception ignored in: <function ResourceTracker.__del__ at 0x103091440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x106359440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x102bad440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x104359440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x104ff5440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x102515440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x1050e5440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/multiprocessing/resource_tracker.py", line 91, in _stop
```

```

File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes
Exception ignored in: <function ResourceTracker.__del__ at 0x102db5440>
Traceback (most recent call last):
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 82, in __del__
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 91, in _stop
  File "/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/
multiprocessing/resource_tracker.py", line 116, in _stop_locked
ChildProcessError: [Errno 10] No child processes

```

5.2 XGBClassifier

```

In [111... pipe_xgb = ImbPipeline(steps=[
    ('prep', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('clf', XGBClassifier(
        use_label_encoder=False,
        eval_metric='logloss',
        random_state=42
    ))
])

scores = cross_val_score(
    pipe_xgb,
    Xtrain, Ytrain,
    cv=5,
    scoring='f1',
    n_jobs=-1
)
print(f"XGBoost 5-Fold CV F1-score: {scores.mean():.3f} ± {scores.std():.3f}")

```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
XGBoost 5-Fold CV F1-score: 0.582 ± 0.013
```

```
In [112... param_grid_xgb = {
    'clf__max_depth':      [3, 5, 7],
    'clf__learning_rate':  [0.01, 0.1, 0.2],
    'clf__n_estimators':   [100, 200],
    'clf__subsample':      [0.8, 1.0],
    'clf__colsample_bytree': [0.8, 1.0]
}

grid_xgb = GridSearchCV(
    pipe_xgb,
    param_grid=param_grid_xgb,
    cv=5,
    scoring='f1',
    n_jobs=-1,
    verbose=1
)
grid_xgb.fit(Xtrain, Ytrain)

print("Best XGBoost Parameters:", grid_xgb.best_params_)
print(f"Best CV F1-score after tuning: {grid_xgb.best_score_:.3f}")
```

Fitting 5 folds for each of 72 candidates, totalling 360 fits

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:53] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:53] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:53] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:53] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:53] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:54] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:54] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:54] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:54] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:57] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:15:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:00] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:03] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:03] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:03] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:03] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:04] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:04] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:09] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:09] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:09] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:09] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:09] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:09] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:09] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:11] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:11] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:13] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:13] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:15] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:16] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:16] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:16] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:19] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:19] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:19] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:20] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:20] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:20] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:20] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:20] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:20] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:21] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:21] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:21] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:21] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:21] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:21] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:21] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:22] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:22] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:22] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:23] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:23] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:23] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:23] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:23] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:23] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:25] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:25] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:25] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:25] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:27] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:27] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:27] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:28] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:28] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:29] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:29] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:29] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:29] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:29] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:30] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:30] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:30] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:31] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:31] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:31] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:31] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:33] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:34] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:34] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:34] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:35] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:35] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:35] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:35] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:35] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:35] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:36] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:36] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:36] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:36] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:36] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:36] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:37] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:37] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:37] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:37] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:38] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:38] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:38] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:38] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:39] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:39] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:39] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:39] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:40] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:40] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:40] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:40] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:41] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:41] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:41] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:41] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:42] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:42] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:42] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:42] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:44] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:44] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:45] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:45] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:45] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:45] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:45] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:46] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:48] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:48] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:48] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:16:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:00] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:00] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:02] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:03] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:03] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:04] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:04] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:04] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:04] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:04] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:10] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:13] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:15] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:22] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:23] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:25] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:25] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:26] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:29] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:30] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:31] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:33] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:39] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:39] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:40] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:41] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:43] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:43] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:44] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:44] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:44] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:45] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:46] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:46] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```



```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:47] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:47] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:47] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:48] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:49] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:50] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:51] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:52] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:53] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:54] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:54] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:55] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:56] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:57] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:57] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:57] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:58] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:17:59] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:00] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:00] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:01] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:05] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:06] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:07] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:08] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:12] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:13] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:13] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:13] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:13] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:14] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:18] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:19] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:24] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.

    bst.update(dtrain, iteration=i, fobj=obj)
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:18:28] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
Best XGBoost Parameters: {'clf__colsample_bytree': 0.8, 'clf__learning_rate': 0.1, 'clf__max_depth': 7, 'clf__n_estimators': 100, 'clf__subsample': 0.8}
```

```
Best CV F1-score after tuning: 0.610
```

```
In [113... final_xgb = ImbPipeline(steps=[
    ('prep', preprocessor),
    ('smote', SMOTE(random_state=42)),
    ('clf', XGBClassifier(
        use_label_encoder=False,
        eval_metric='logloss',
        learning_rate=0.1,
        max_depth=7,
        n_estimators=100,
        subsample=0.8,
        colsample_bytree=0.8,
        random_state=42
    ))
])

final_xgb.fit(Xtrain, Ytrain)

y_pred = final_xgb.predict(Xtest)
y_proba = final_xgb.predict_proba(Xtest)[:, 1]
```

```
/Library/Frameworks/Python.framework/Versions/3.13/lib/python3.13/site-packages/xgboost/training.py:183: UserWarning: [20:19:39] WARNING: /Users/runner/work/xgboost/xgboost/src/learner.cc:738:
Parameters: { "use_label_encoder" } are not used.
```

```
bst.update(dtrain, iteration=i, fobj=obj)
```

```
In [117... print("          === Final XGBoost Classification Report ===")
print(classification_report(Ytest, y_pred, target_names=['No', 'Yes']))
print(f"ROC AUC : {roc_auc_score(Ytest, y_proba):.2f}")

fpr, tpr, _ = roc_curve(Ytest, y_proba)
plt.figure(figsize=(8,6))
plt.plot(fpr, tpr, label=f'Final XGBoost (AUC = {roc_auc_score(Ytest, y_p
plt.plot([0,1],[0,1], '--', color='red')
plt.title('ROC Curve - Final XGBoost Model')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.legend(loc='lower right')
plt.grid(True)
plt.show()
```



```

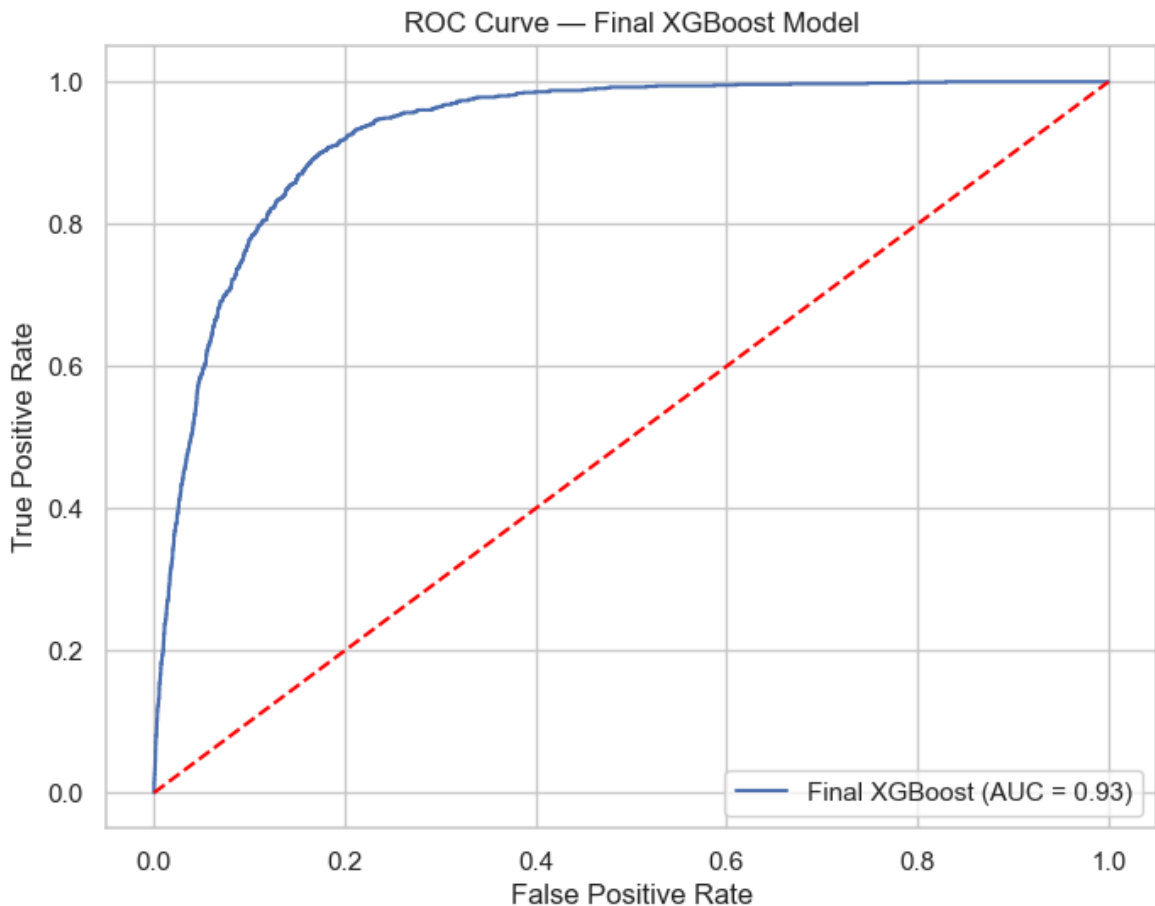
=== Final XGBoost Classification Report ===
      precision    recall  f1-score   support

   No       0.96       0.93       0.94       7985
   Yes       0.57       0.67       0.62       1058

 accuracy          0.90          9043
  macro avg       0.76       0.80       0.78          9043
 weighted avg     0.91       0.90       0.91          9043

```

ROC AUC : 0.93



Comparison

Metric	CatBoost	XGBoost
Accuracy	0.90	0.90
Precision (Yes)	0.57	0.57
Recall (Yes)	0.69	0.67
F1-score (Yes)	0.63	0.62
ROC AUC	0.928	0.929
5-Fold CV F1	0.596 ± 0.012	0.582 ± 0.013

Overall, both tuned CatBoost and XGBoost deliver very strong performance on this imbalanced classification task, achieving nearly identical accuracy (90%) and AUC (≈ 0.93). CatBoost pulls ahead with a slightly better F1-score (0.63 vs. 0.62) and

higher recall (0.69 vs. 0.67) on the positive ("yes") class, suggesting it strikes a marginally better balance between catching true positives and avoiding false alarms. XGBoost, in turn, edges out by a hair on AUC and often trains a bit faster, making it attractive when throughput is critical. If your highest priority is maximizing the F1-score on the minority class and you value native handling of categorical features, CatBoost is the way to go; if you need the smallest possible latency or align with an existing XGBoost-based pipeline, XGBoost remains an excellent choice.